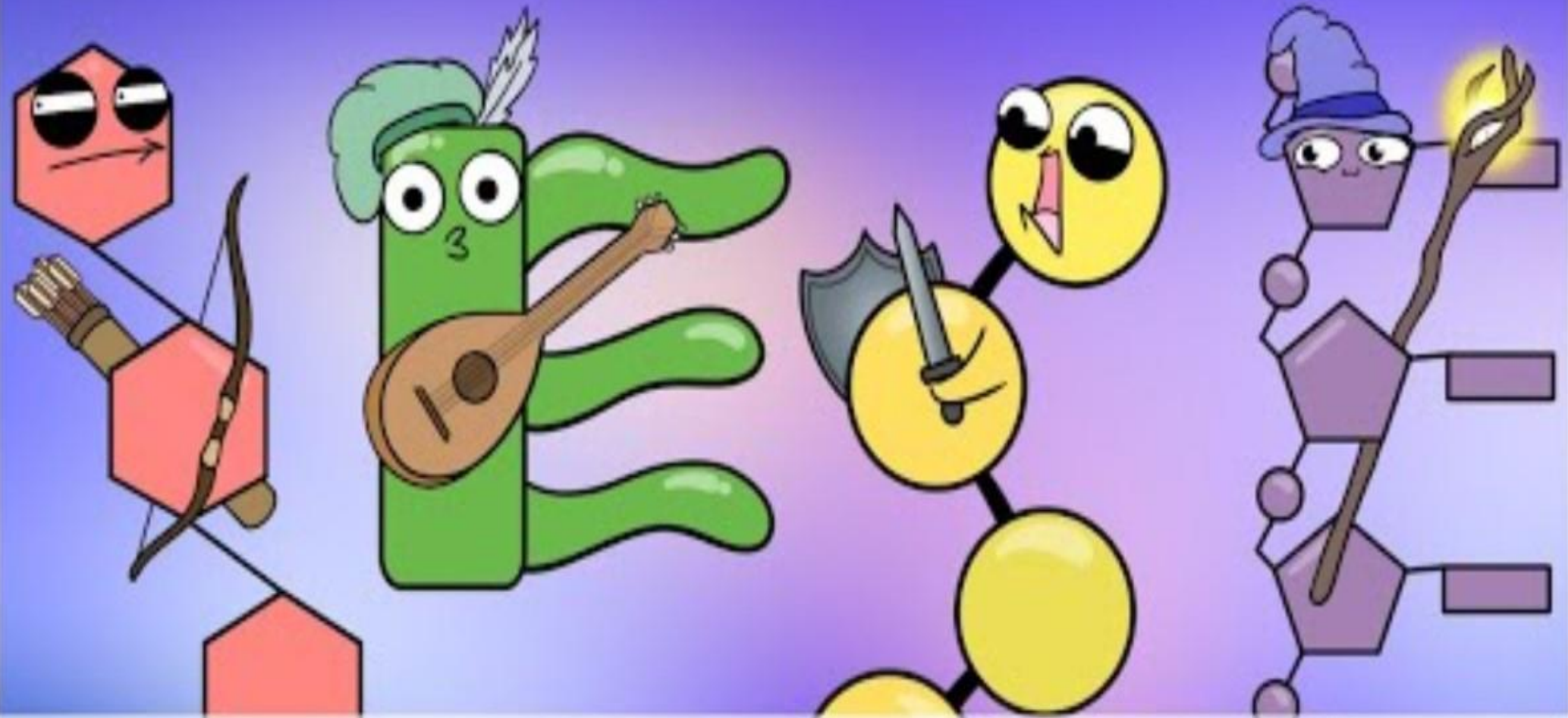
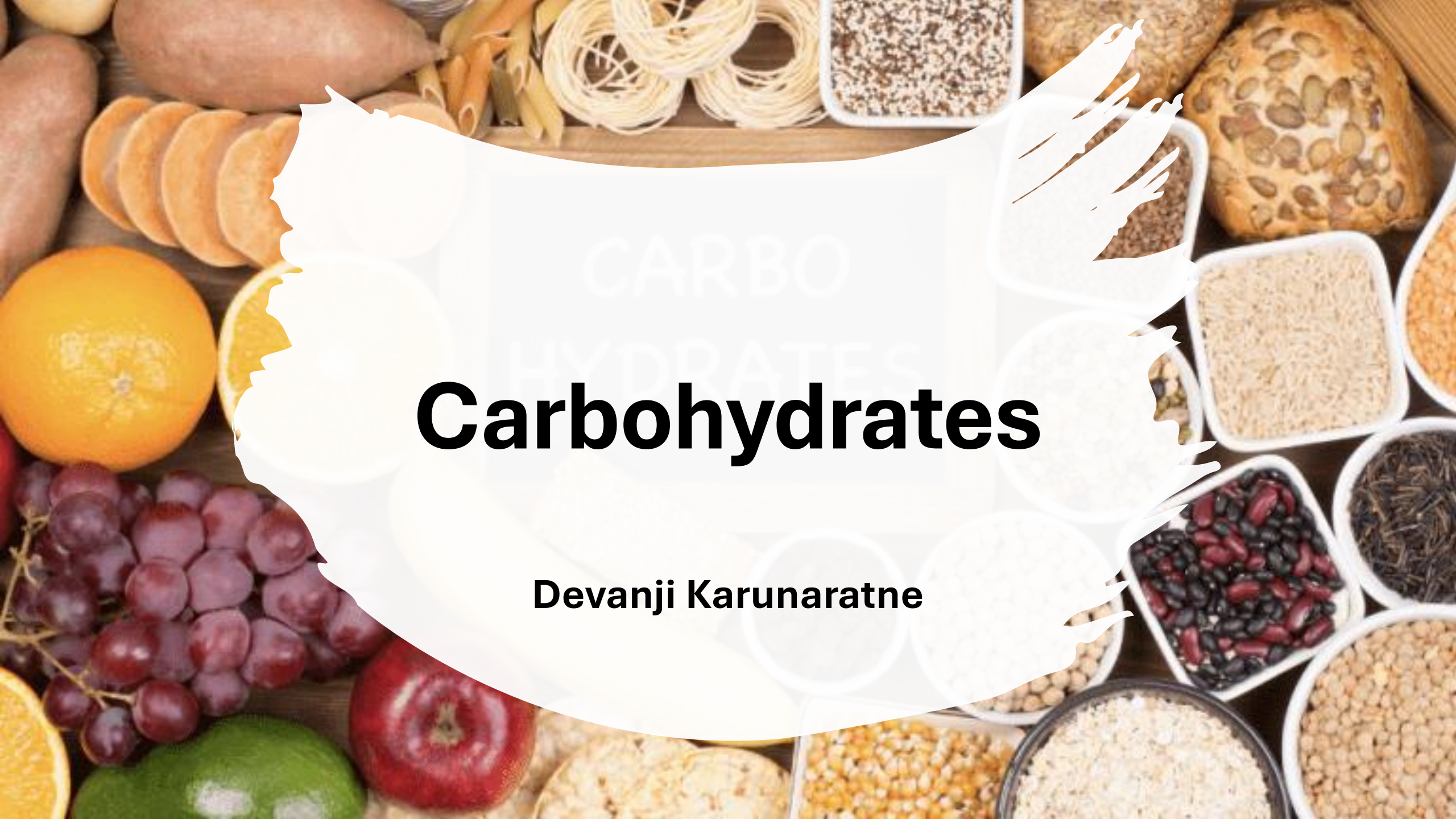




Biomolecules (Updated 2023)

with the Amoeba Sisters



Carbohydrates

Devanji Karunaratne

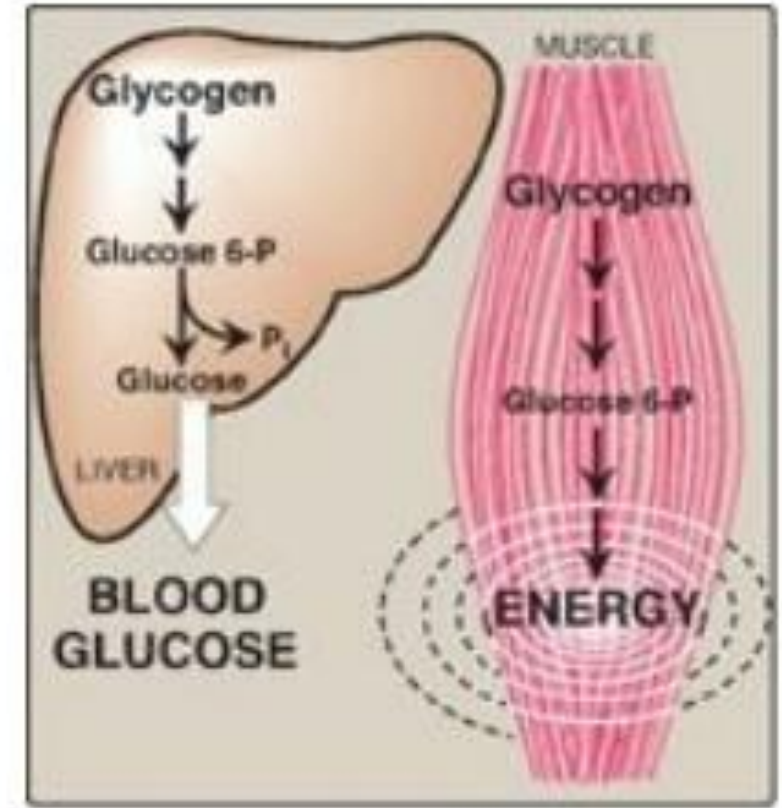
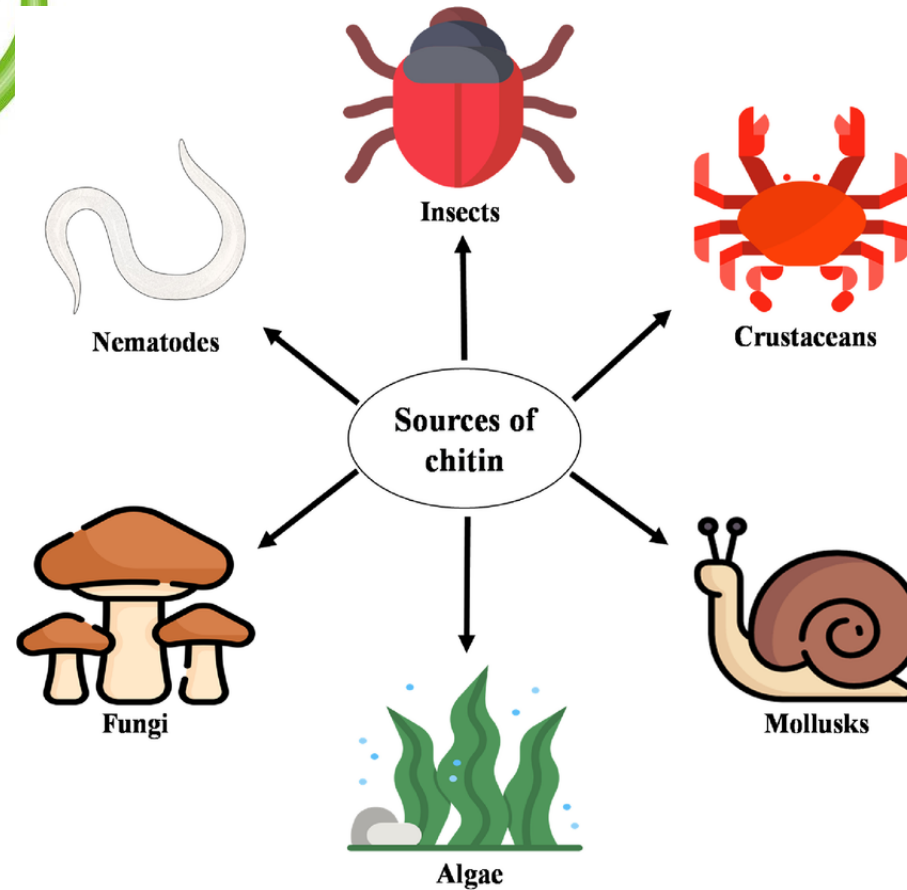
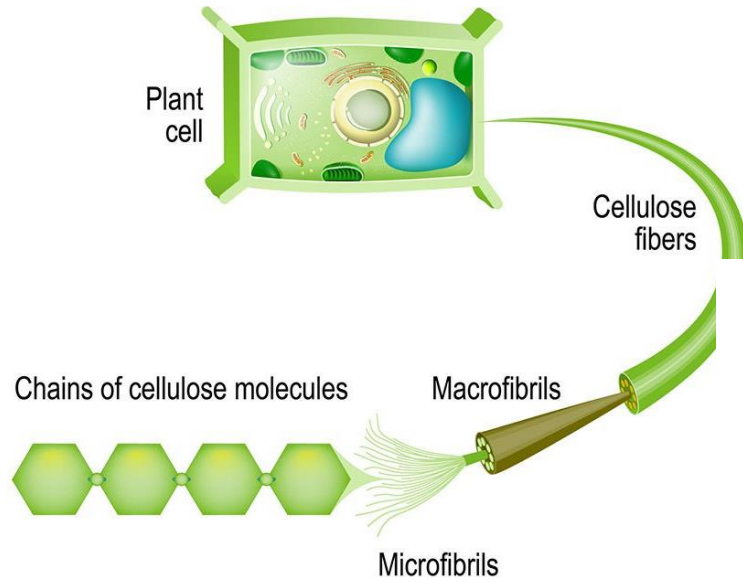
What are Carbohydrates?

- Chemically, carbohydrates are defined as polyhydroxy aldehydes or ketones.
- Their molecular structure contains multiple hydroxyl (-OH) groups and at least one carbonyl group (either an aldehyde, -CHO, or a ketone, -C=O).
- Their empirical formula is often $C_n(H_2O)_n$.

Why are Carbohydrates so important in Biology?

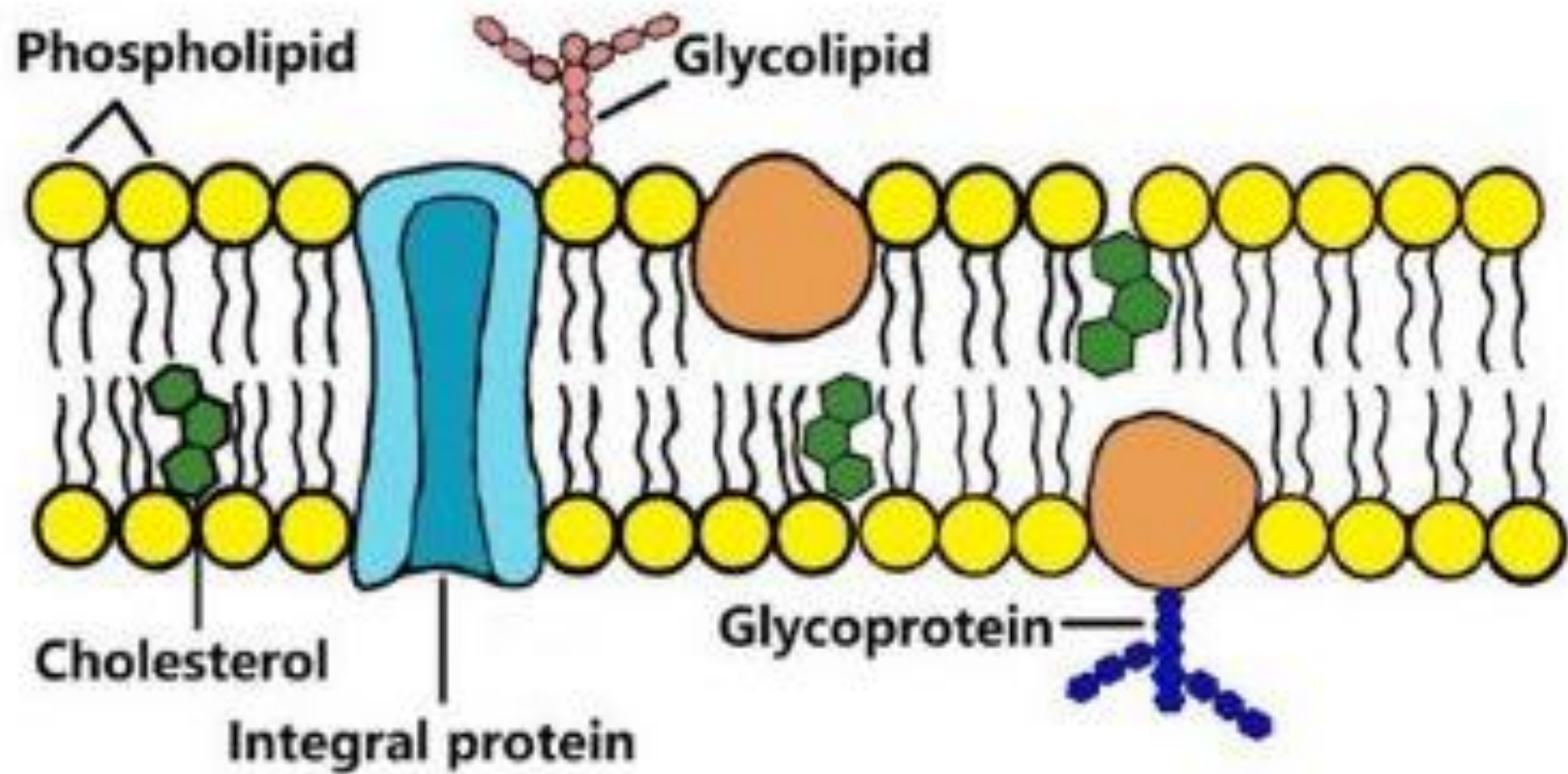
- **Primary Energy Source:** Glucose is the immediate fuel source for most living cells, powering vital processes from muscle contraction to nerve impulses.
- **Energy Storage:** Plants store excess glucose as **starch**, and animals (including humans) store it as **glycogen** (primarily in the liver and muscles). These are readily accessible energy reserves.
- **Structural Components:** Carbohydrates provide structural integrity. The rigid cell walls of plants are composed of **cellulose**, a carbohydrate that gives plants their upright form. The tough exoskeletons of insects and crustaceans, and the cell walls of fungi, are made of another carbohydrate called **chitin**.

CELLULOSE



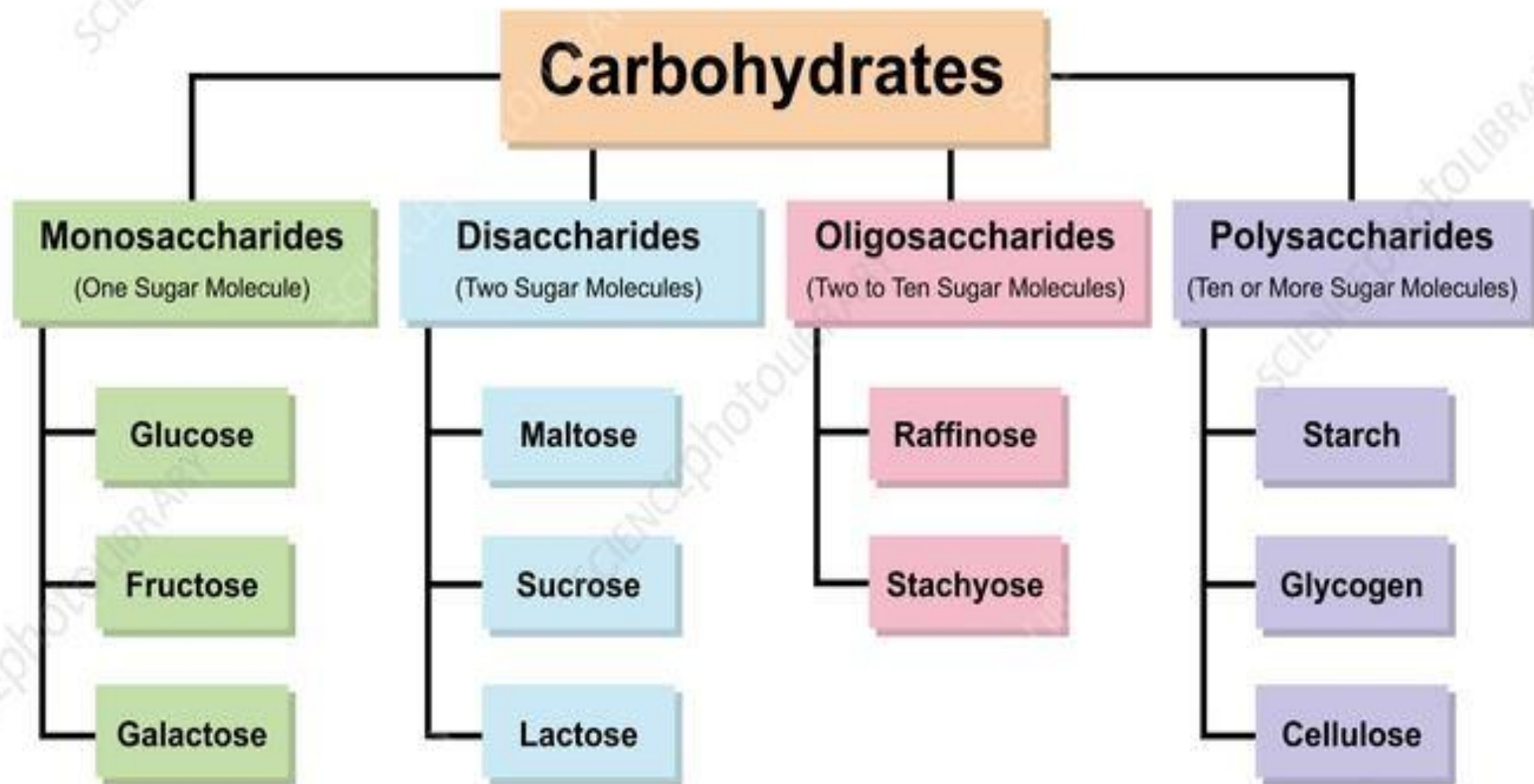
Why are Carbohydrates So Important in Biology?

- **Cell Recognition and Signaling:** On the surface of cells, carbohydrates are often found linked to proteins (**glycoproteins**) or lipids (**glycolipids**). These complex carbohydrate structures, allow cells to recognize each other, and participate in crucial communication processes (e.g., in immune responses, blood typing).
- **Precursors for Other Biomolecules:** Carbohydrates can also be converted into other important biomolecules, such as amino acids and lipids.



Classification of Carbohydrates

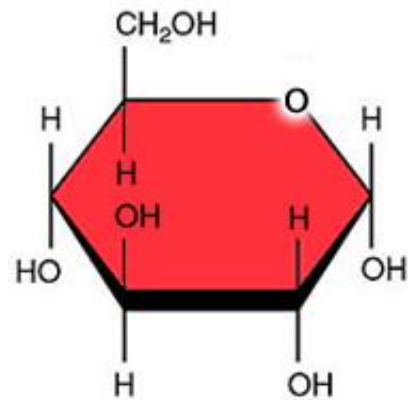
- **Monosaccharides:** "Mono" means one. These are the simplest sugars, single polyhydroxy aldehyde or ketone units. They cannot be broken down into smaller sugar units by hydrolysis. **They are the fundamental building blocks.**
- **Disaccharides:** "Di" means two. These are composed of **two monosaccharide** units linked together. They can be hydrolyzed (broken down with water) into their two constituent monosaccharides.
- **Oligosaccharides:** "Oligo" means few (typically **3-10 sugar units**). These often play roles in cell recognition.
- **Polysaccharides:** "Poly" means many. These are large polymers made of hundreds or even **thousands of monosaccharide** units. They serve major roles in energy storage and structural support.



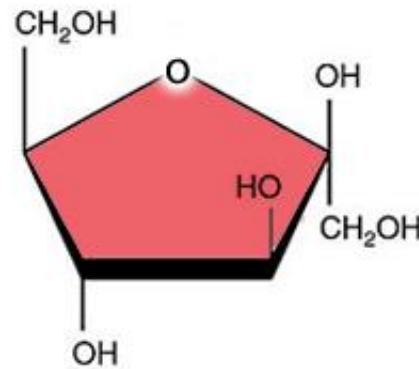
Monosaccharides

Monosaccharides are the simplest forms of carbohydrates. They are typically colorless, crystalline solids, freely soluble in water, and most have a sweet taste.

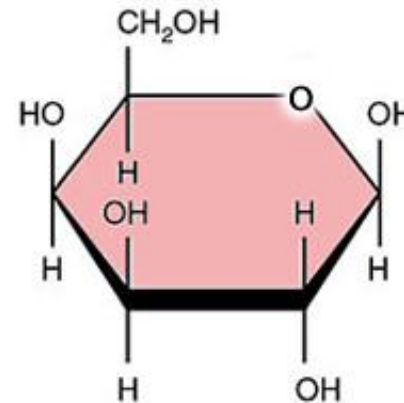
Monosaccharides



Glucose



Fructose



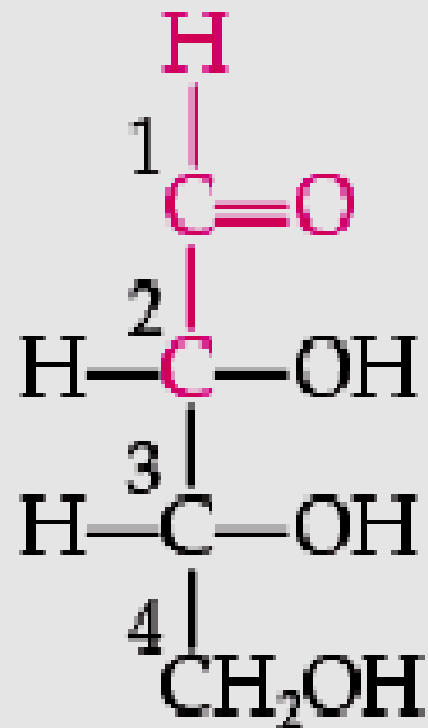
Galactose

Structure and Classification of Monosaccharides

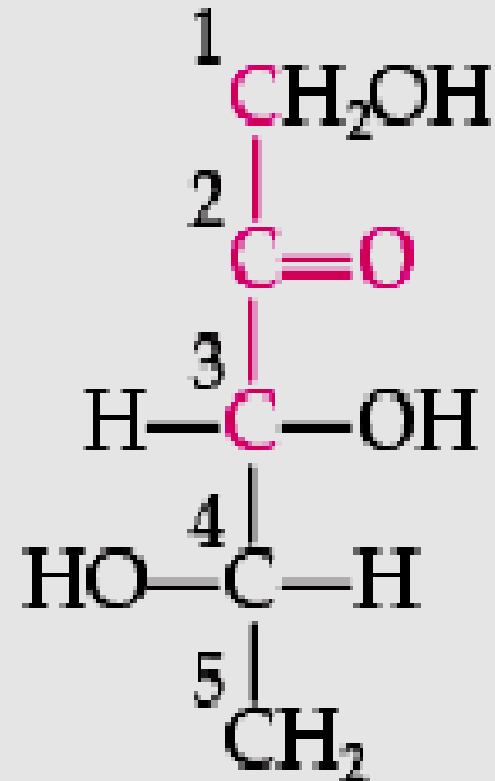
Based on the Nature of the Carbonyl Group

- **Aldoses:** These contain an aldehyde group (-CHO) at one end of the carbon chain. The general formula is R-CHO .
*Example: **Glucose*** (an aldohexose). This is the most prevalent monosaccharide in living systems, central to energy metabolism.
- **Ketoses:** These contain a ketone group (-C=O), typically located at the second carbon atom (C_2) from the end of the chain. The general formula is R-CO-R' .
*Example: **Fructose*** (a ketohexose). Found in fruits, honey, and high-fructose corn syrup.

Aldehyde
functional
group

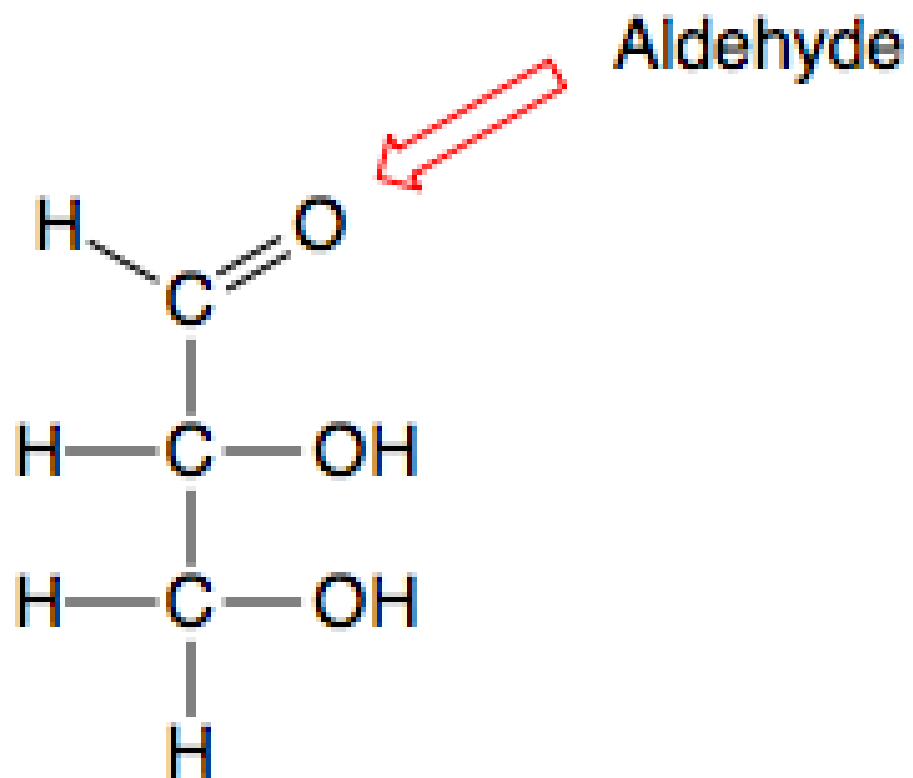


An aldose

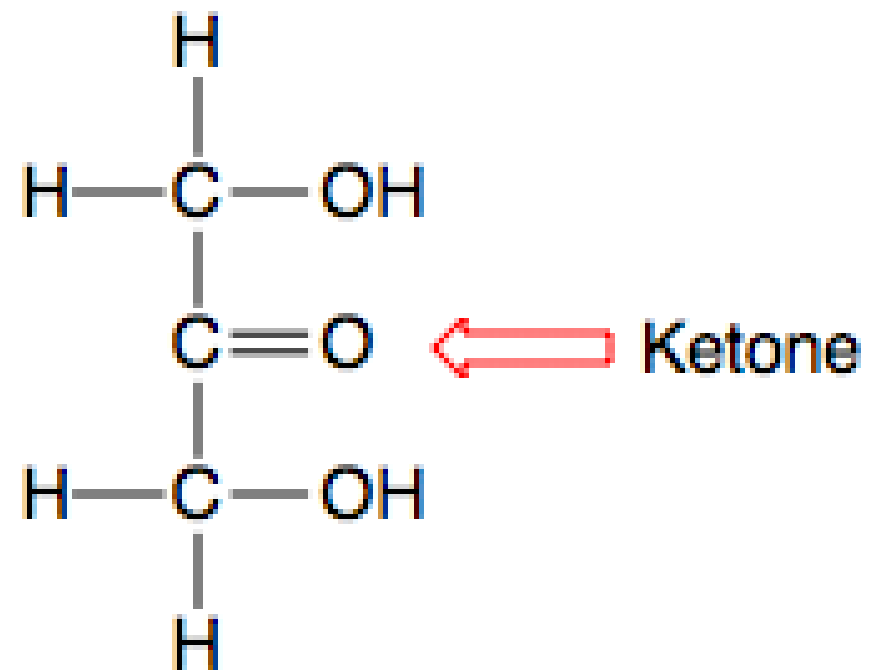


A ketose

Ketone
functional
group



An aldose



A ketose

Structure and Classification of Monosaccharides

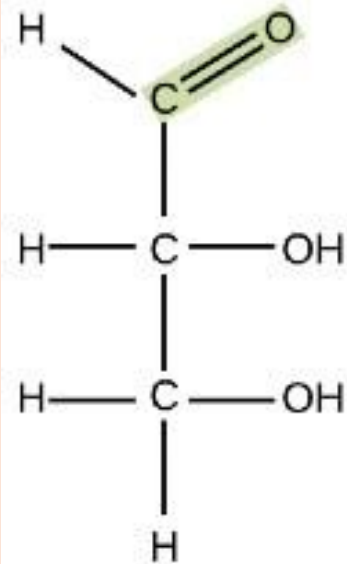
Based on the Number of Carbon Atoms

- **Trioses:** 3 carbons (e.g., **Glyceraldehyde** – key intermediates in glycolysis, the pathway that breaks down glucose for energy).
- **Tetroses:** 4 carbons (e.g., **Erythrose**).
- **Pentoses:** 5 carbons (e.g., **Ribose** found in RNA and ATP; **Deoxyribose** found in DNA. These are crucial components of nucleic acids, which carry genetic information).
- **Hexoses:** 6 carbons (e.g., **Glucose**, **Fructose**, **Galactose** (a component of milk sugar), **Mannose**). These are the most common and metabolically important monosaccharides.

Monosaccharides

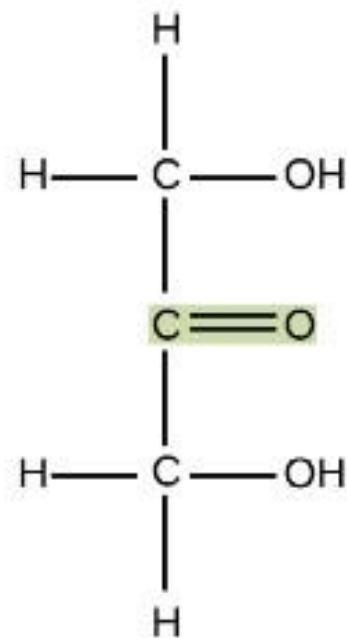
aldose

glyceraldehyde



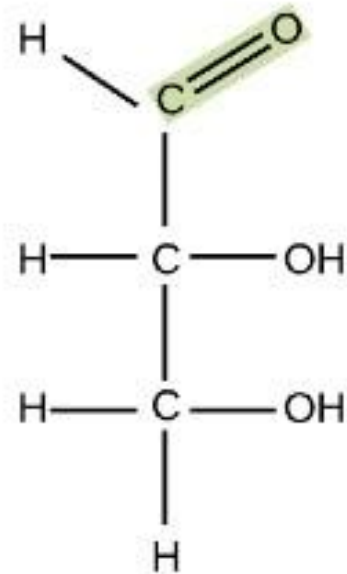
ketose

dihydroxyacetone



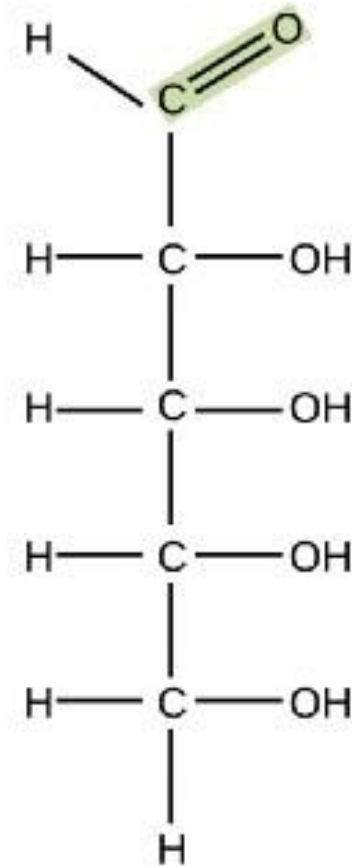
triose

glyceraldehyde



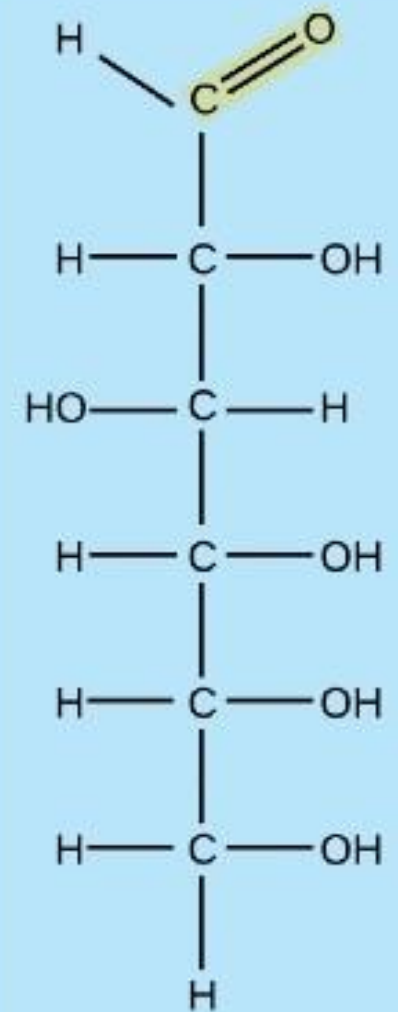
pentose

ribose



hexose

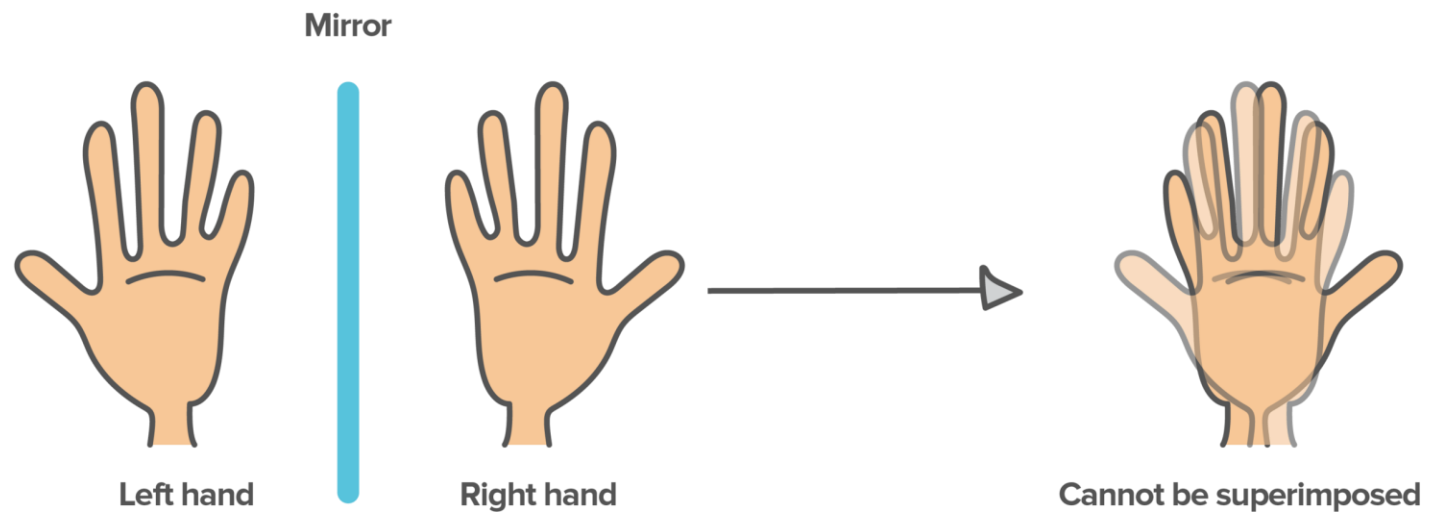
glucose



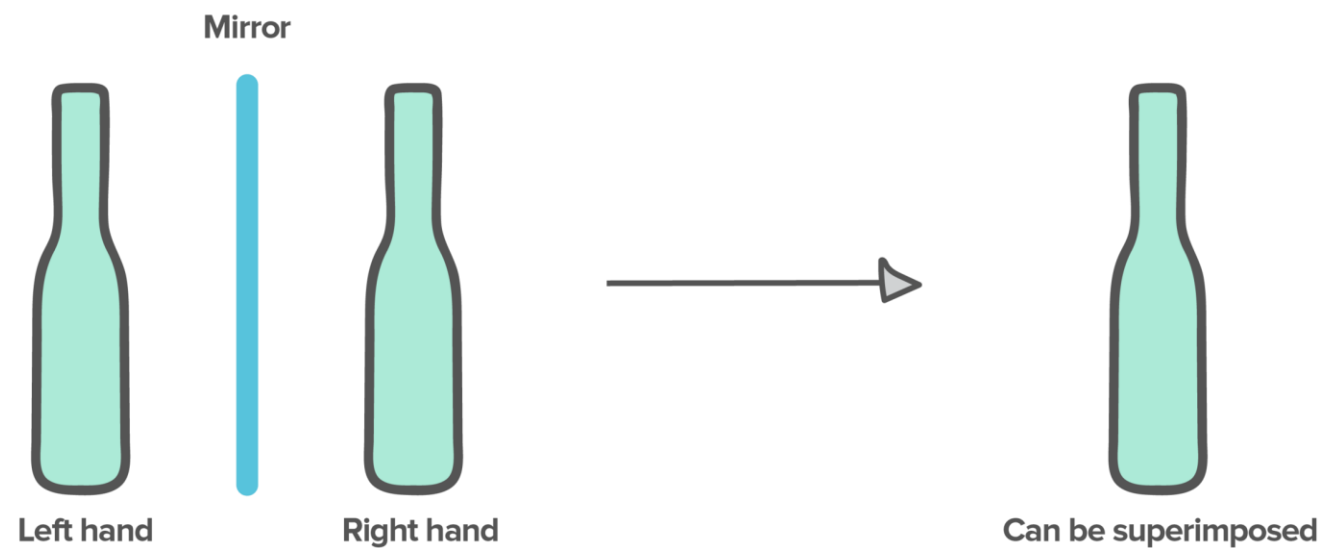
Chirality and Stereoisomers

- Monosaccharides are rich in **chiral centers** (also called asymmetric carbon atoms).
- A chiral carbon is one bonded to four *different* groups.
- This property leads to the existence of stereoisomers – molecules with the same chemical formula but different spatial arrangements of atoms.

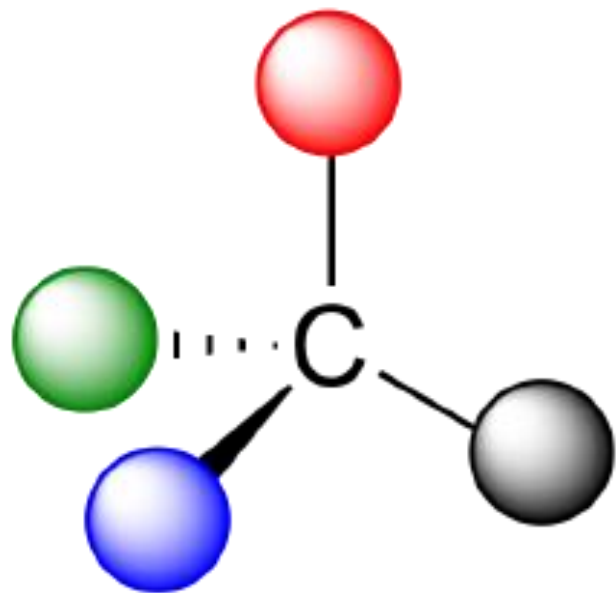
CHIRAL OBJECTS



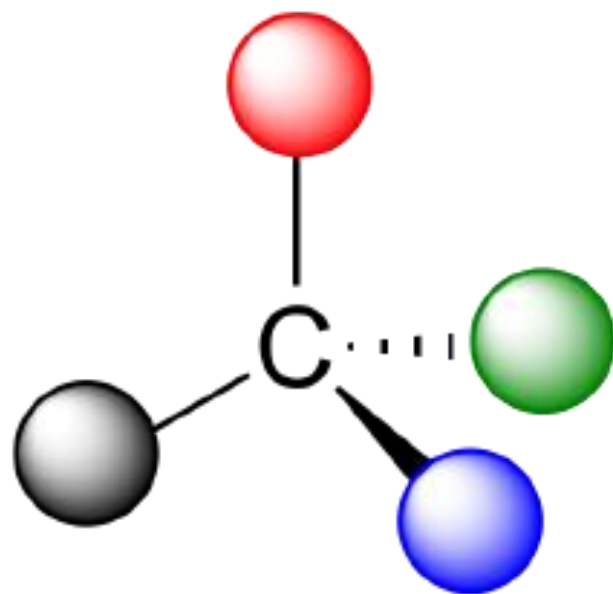
ACHIRAL OBJECTS



mirror plane



A



B

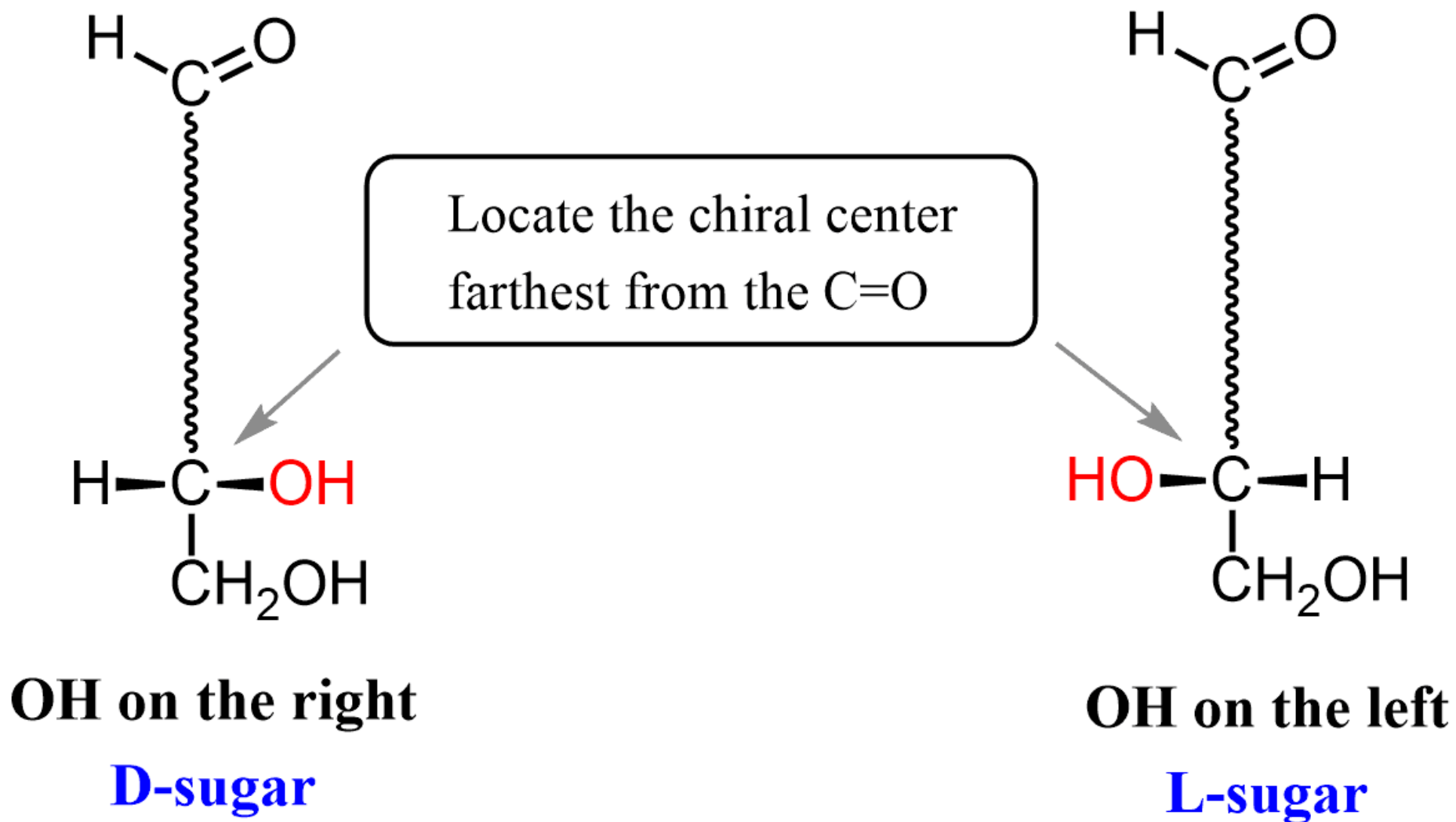
Chirality and Stereoisomers

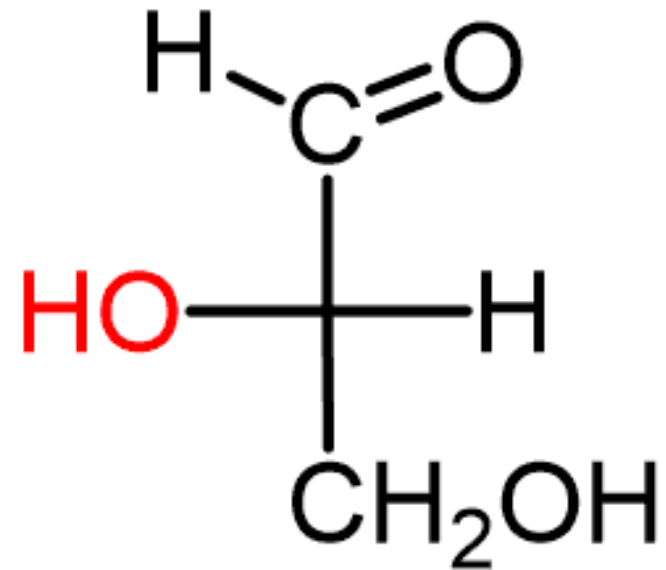
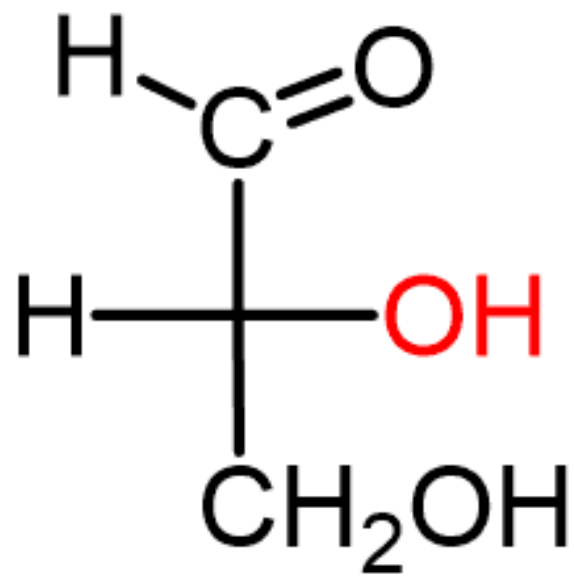
D- and L- Designation: This convention relates to the configuration of the chiral carbon atom that is *furthest* from the carbonyl group (this is called the reference carbon).

- Most naturally occurring sugars in biological systems are **D-isomers**.
- In the Fischer projection (a 2D representation), the -OH group on the lowest chiral carbon is on the *right* for a D-sugar.

Example: **D-Glyceraldehyde** has the -OH on C2 on the right. **L-Glyceraldehyde** has it on the left.

D and L Configuration of Carbohydrates

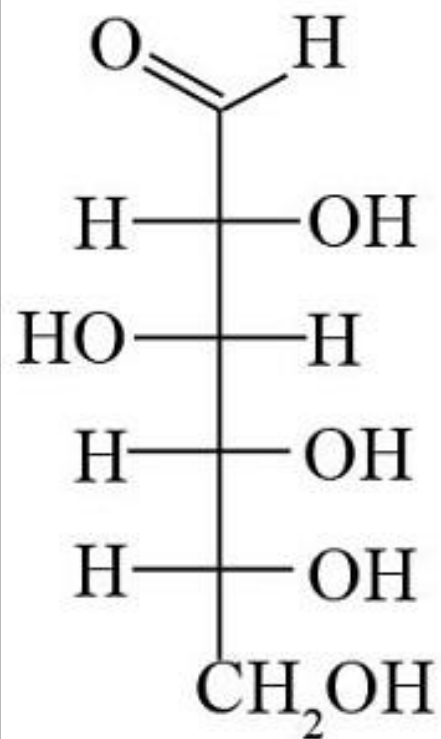




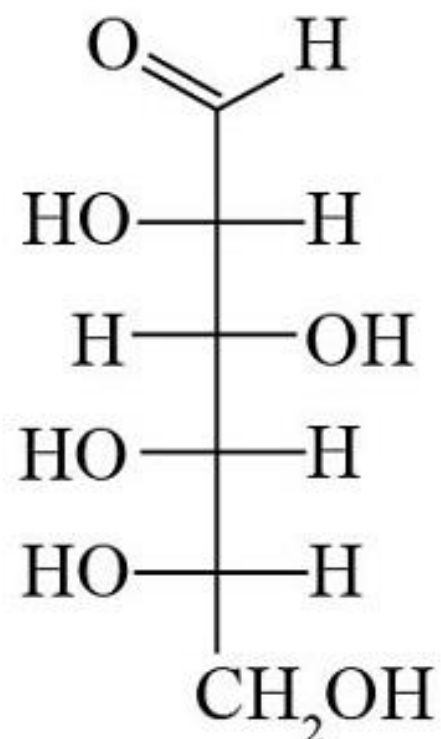
enantiomers

D-glyceraldehyde

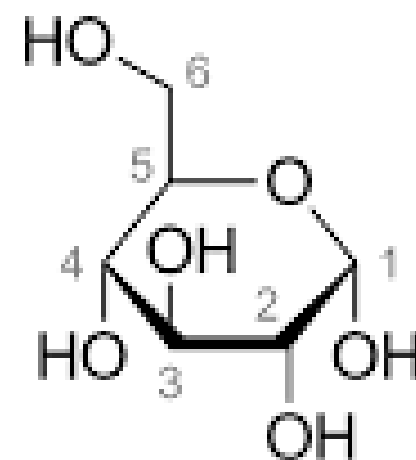
L-glyceraldehyde



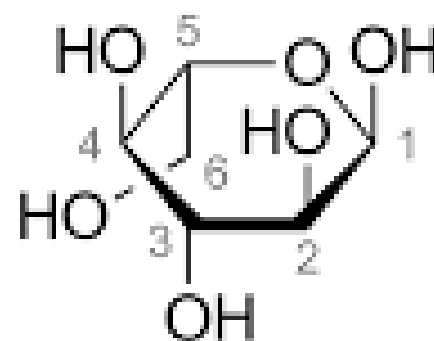
D-Glucose



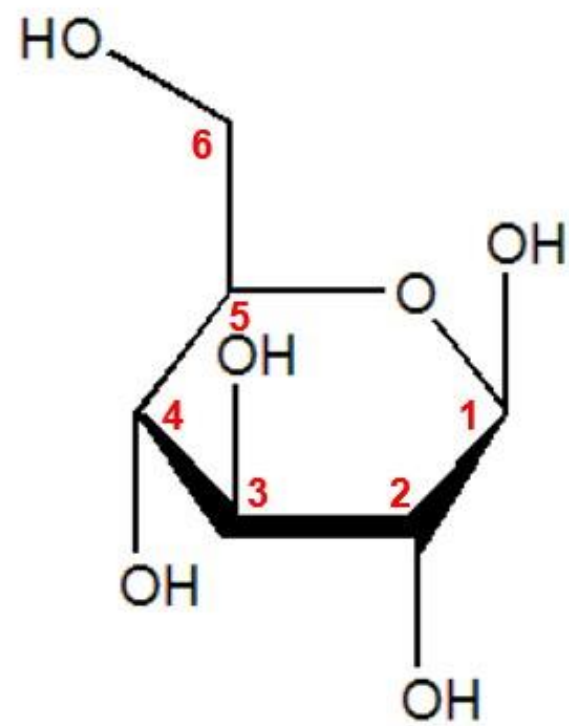
L-Glucose



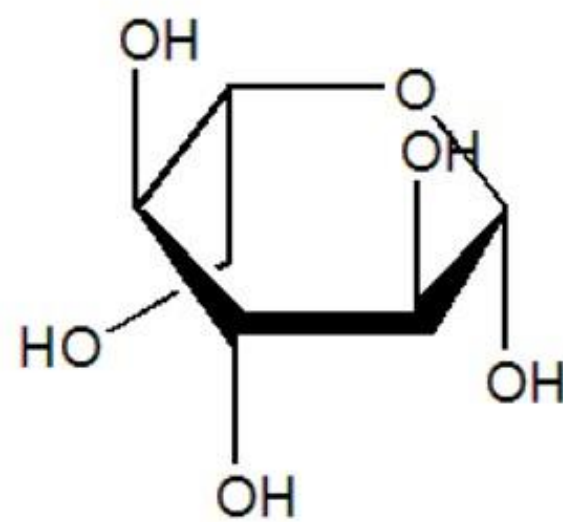
α -D-glucopyranose



α -L-glucopyranose



β -D-glucose



β -L-glucose



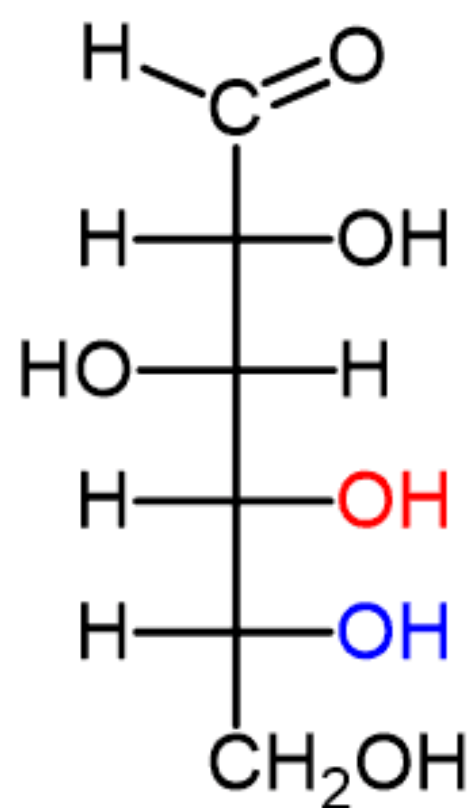
DIFFERENCE BETWEEN
D-GLUCOSE AND **L-GLUCOSE**

Chirality and Stereoisomers

Epimers: These are specific types of diastereomers (stereoisomers that are not mirror images) that differ in configuration at *only one* chiral carbon.

- *Example:* **D-Glucose and D-Galactose** are C4 epimers (they differ only in the arrangement of -OH at carbon 4).
- *Example:* **D-Glucose and D-Mannose** are C2 epimers (they differ only at carbon 2). This subtle difference has significant biological implications for enzyme recognition.

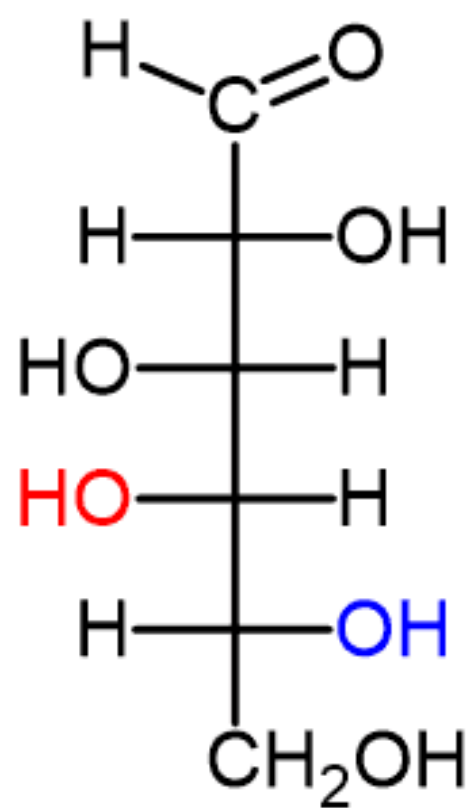
D-glucose and D-galactose are epimeric at carbon-4



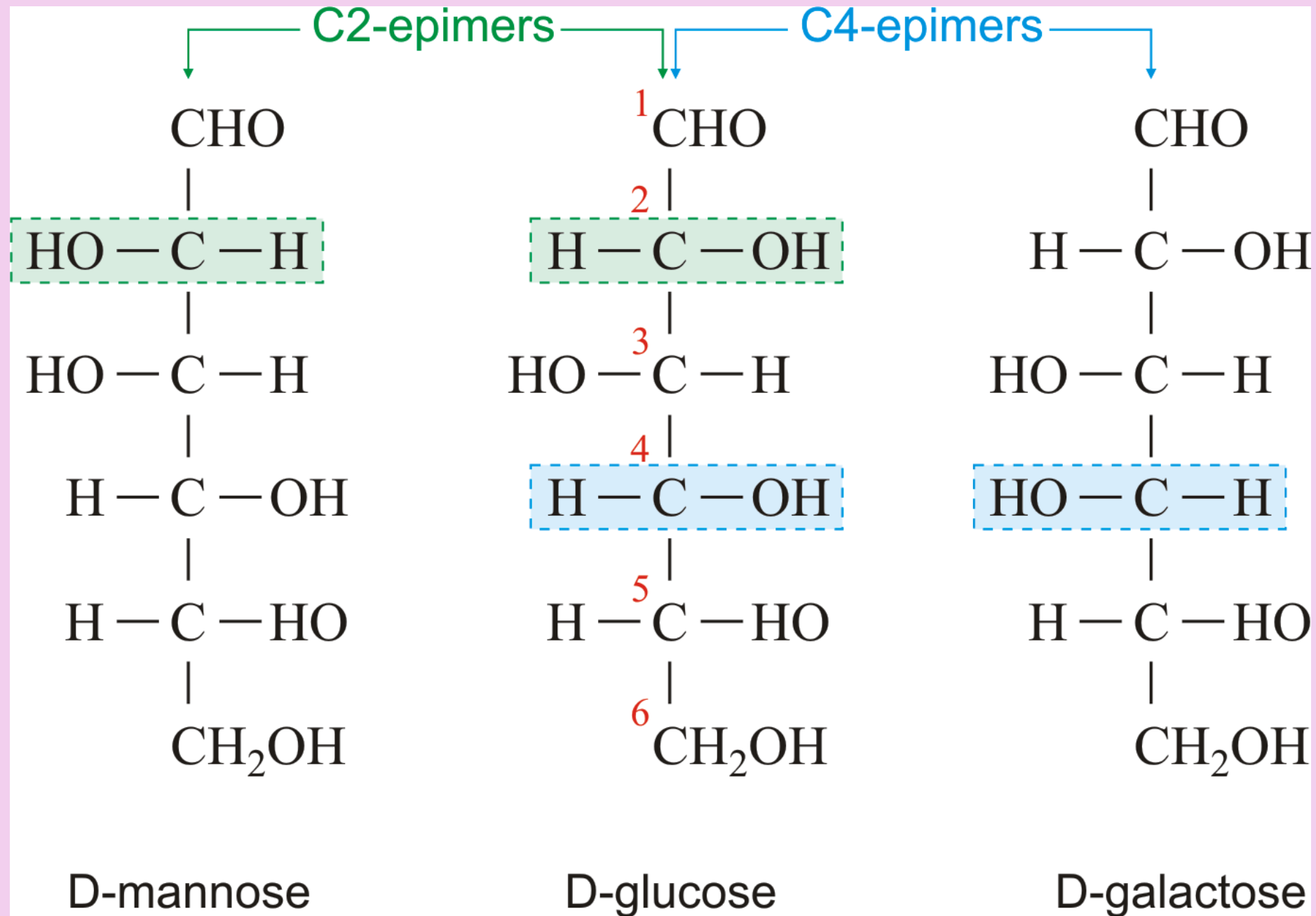
D-Glucose

Epimers

Only one chiral
center with
different
configuration



D-Galactose



Cyclic Structures

In aqueous solutions, which is the environment inside cells, monosaccharides with five or more carbons do *not* primarily exist as straight chains. Instead, they spontaneously cyclize to form rings.

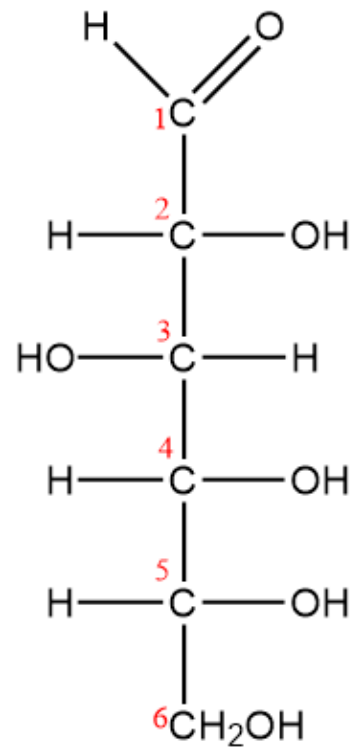
How Ring Formation Occurs: The aldehyde or ketone group ($\text{C}=\text{O}$) on one end of the molecule reacts with a hydroxyl ($-\text{OH}$) group on another carbon within the *same* molecule.

- Aldoses form **hemiacetals**.
- Ketoses form **hemiketals**.

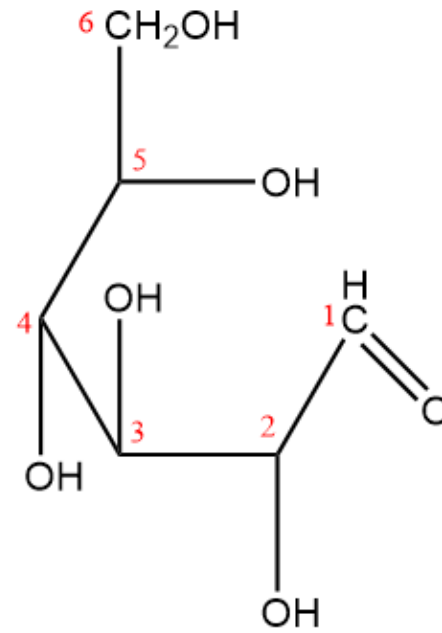
The cyclization
of glucose to a
Haworth projection

Common Ring Sizes

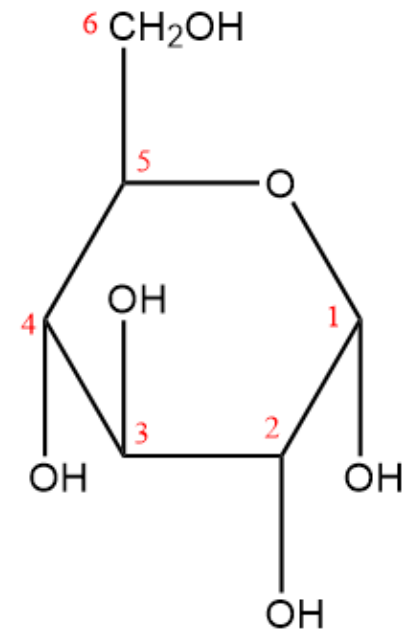
- **Pyranose Ring:** A stable six-membered ring that contains five carbon atoms and one oxygen atom (e.g., α -D-glucopyranose, which is the common cyclic form of glucose).
- **Furanose Ring:** A five-membered ring that contains four carbon atoms and one oxygen atom (e.g., β -D-fructofuranose, a common cyclic form of fructose).



a. Straight Chain Monosaccharide



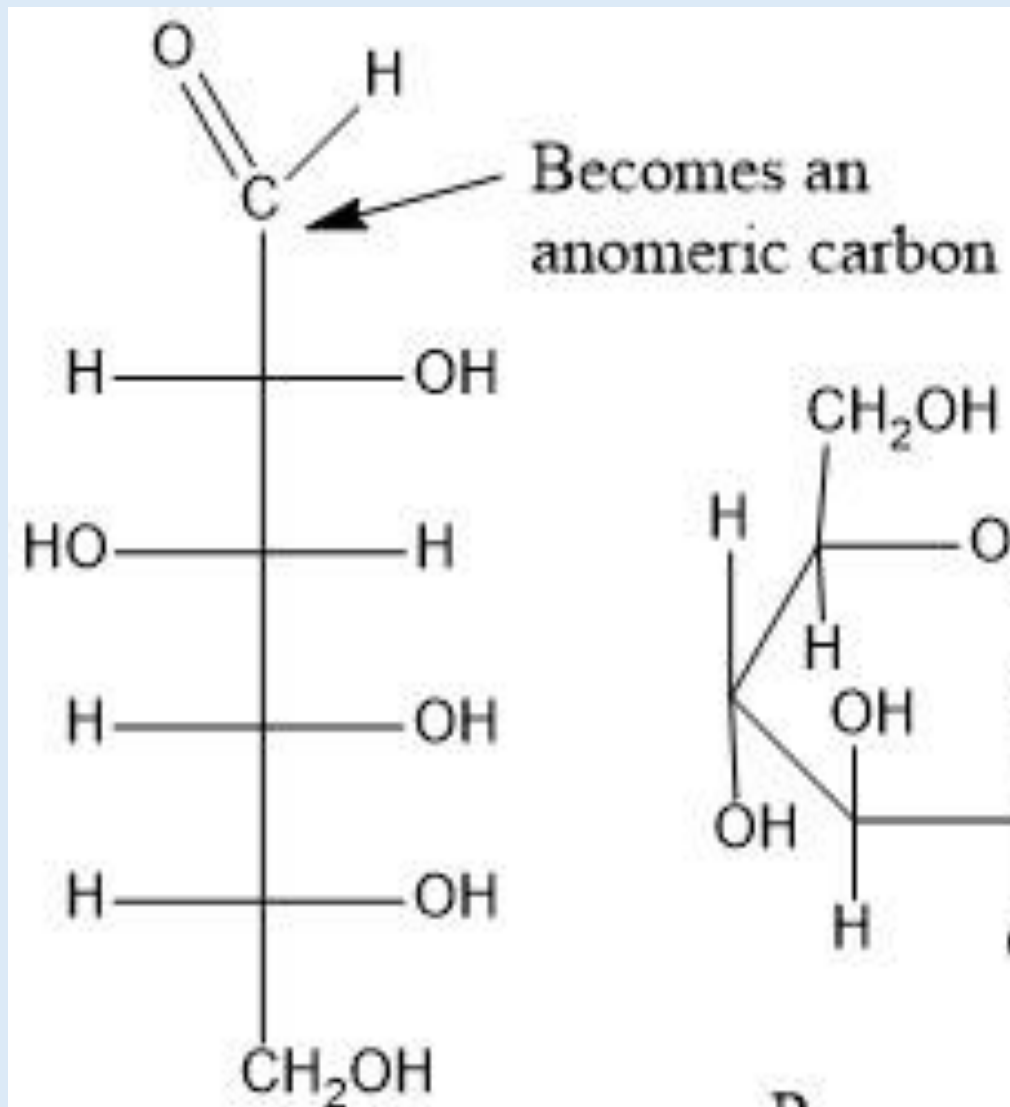
b. Open form of Monosaccharide



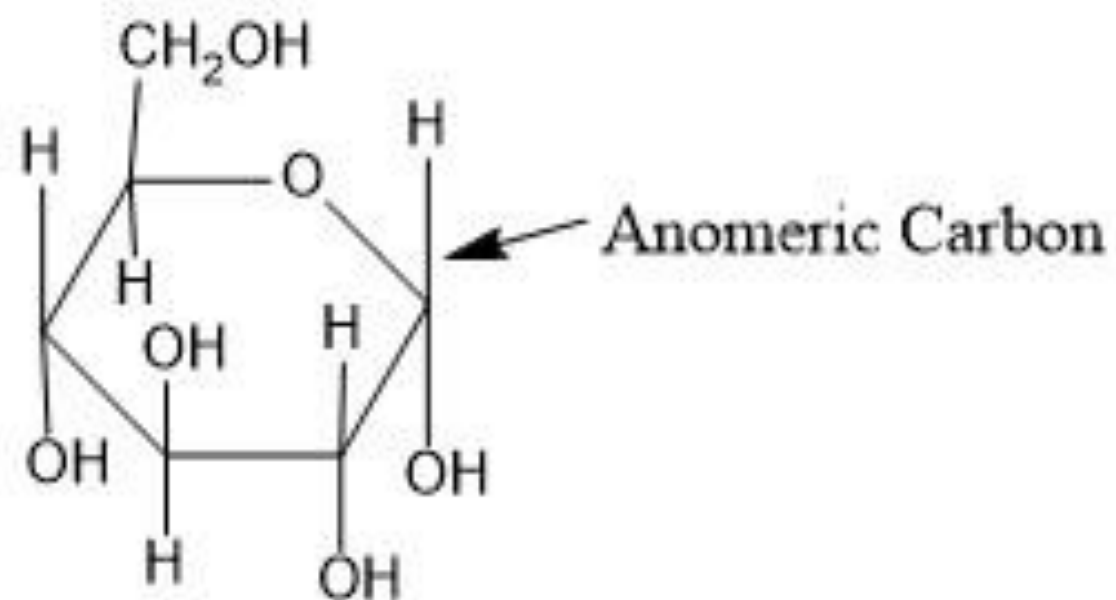
c. Cyclic Monosaccharide

The Anomeric Carbon

- This ring closure creates a *new* chiral center at the carbon atom that was originally the carbonyl carbon (C1 for aldoses, C2 for ketoses).
- This newly chiral carbon is called the **anomeric carbon**.



D-Glucose

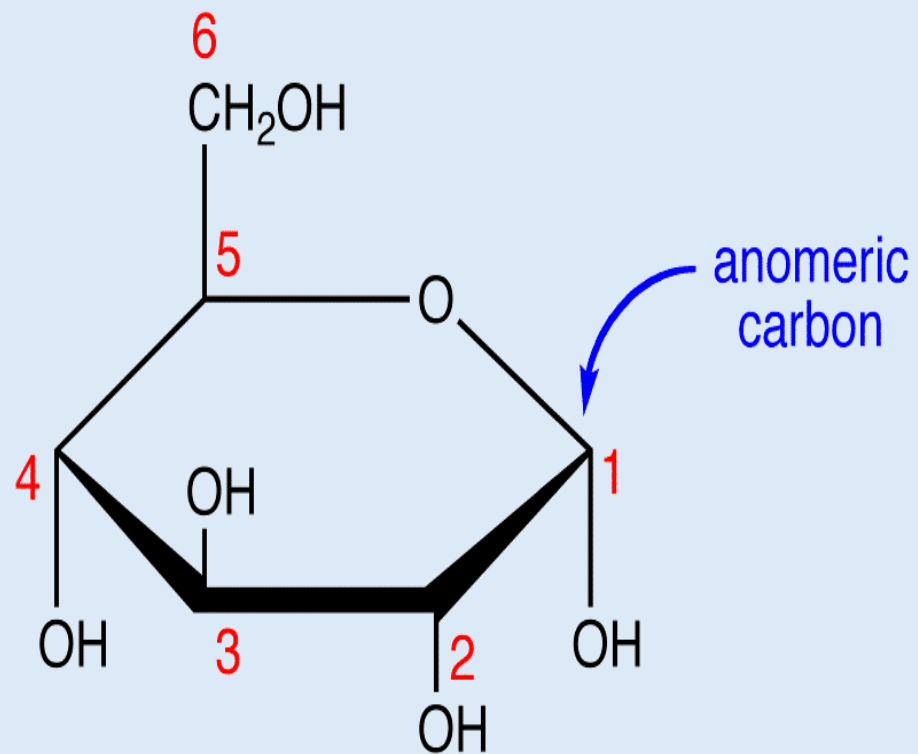


Pyranose form of
D-Glucose

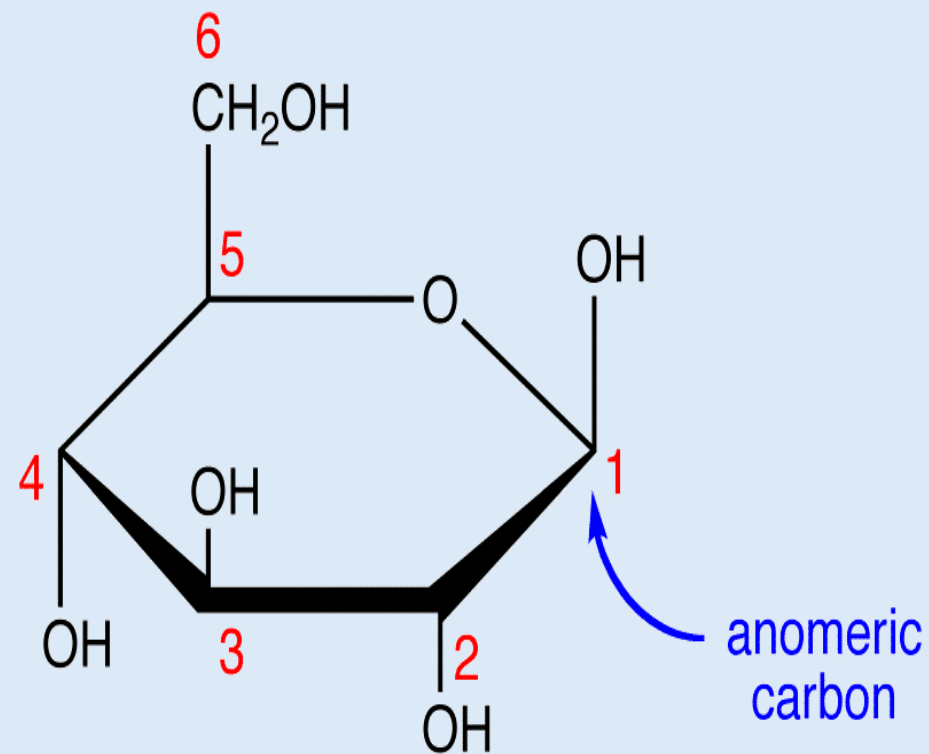
Anomers (α and β)

The anomeric carbon can exist in two distinct stereoisomeric forms, called anomers, depending on the orientation of its newly formed hydroxyl group:

- **α -anomer:** The -OH group on the anomeric carbon is on the *opposite side* of the ring from the -CH₂OH group at the highest numbered chiral carbon (for D-sugars).
- **β -anomer:** The -OH group on the anomeric carbon is on the *same side* of the ring from the -CH₂OH group (for D-sugars).



α -D-glucopyranose



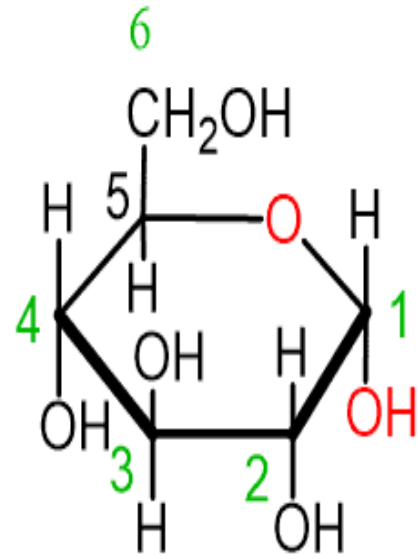
β -D-glucopyranose

Mutarotation: The Dynamic Equilibrium in Solution

- If you dissolve a pure crystal of α -D-glucose in water, you'll observe that its optical rotation (how it rotates plane-polarized light) slowly changes until it reaches a stable, equilibrium value.
- This process is known as **mutarotation**.

What's Happening? In solution, the α -anomer, β -anomer, and the open-chain form of the sugar are in a dynamic equilibrium, constantly interconverting via the transient open-chain intermediate.

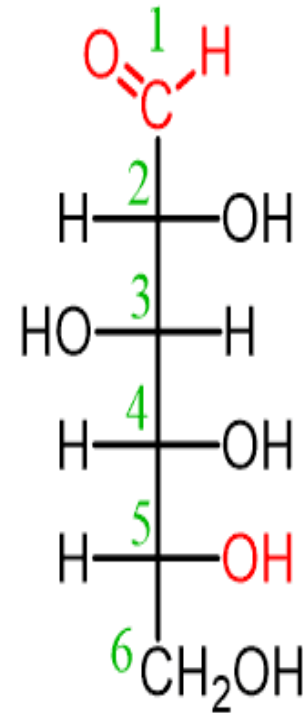
This means a solution of "glucose" is actually a mixture of these forms.



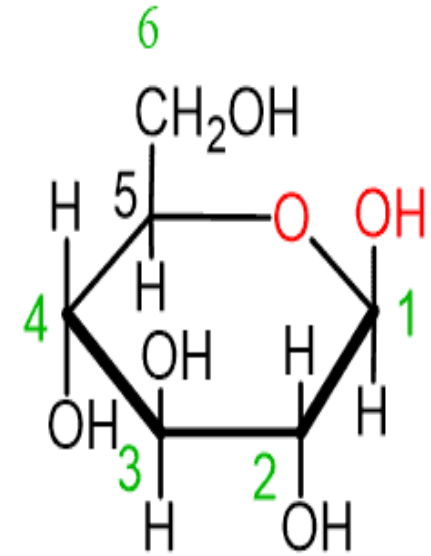
α -D-(+)-glucose

(mp 146 °C, $[\alpha] = +112$)

37%



open chain form of
D-(+)-glucose



β -D-(+)-glucose

(mp 150 °C, $[\alpha] = +18.7$)

63%

Key Reactions of Monosaccharides

Oxidation Reactions

- The free aldehyde group of an aldose (or the ketone group of a ketose, which can isomerize to an aldehyde in alkaline conditions) can be readily oxidized.
- Sugars that have a free aldehyde or ketone group (or can form one in solution) and can therefore be oxidized are called **reducing sugars**.
- This property is the basis for classic laboratory tests and modern glucose meters.

Example: Benedict's Test

1 mL
Benedict's
Reagent



2 - 3
drops
sample



Heat
3 minutes



Amount of reducing sugars

None



Blue
Solution

High



Red
Precipitate

Medium



Orange
Precipitate

Low



Green
Precipitate

Key Reactions of Monosaccharides

Reduction Reactions

- The carbonyl group (either aldehyde or ketone) of a monosaccharide can be reduced to a hydroxyl group, yielding compounds called **sugar alcohols** or **alditols**.

Example

- Reduction of glucose yields **sorbitol**.
- Reduction of fructose yields a mixture of **sorbitol** and **mannitol**.
- **Applications:** Sorbitol is widely used as a sugar substitute in "sugar-free" foods (e.g., chewing gum, diet candies) because it is metabolized more slowly than glucose and does not promote tooth decay.

Key Reactions of Monosaccharides

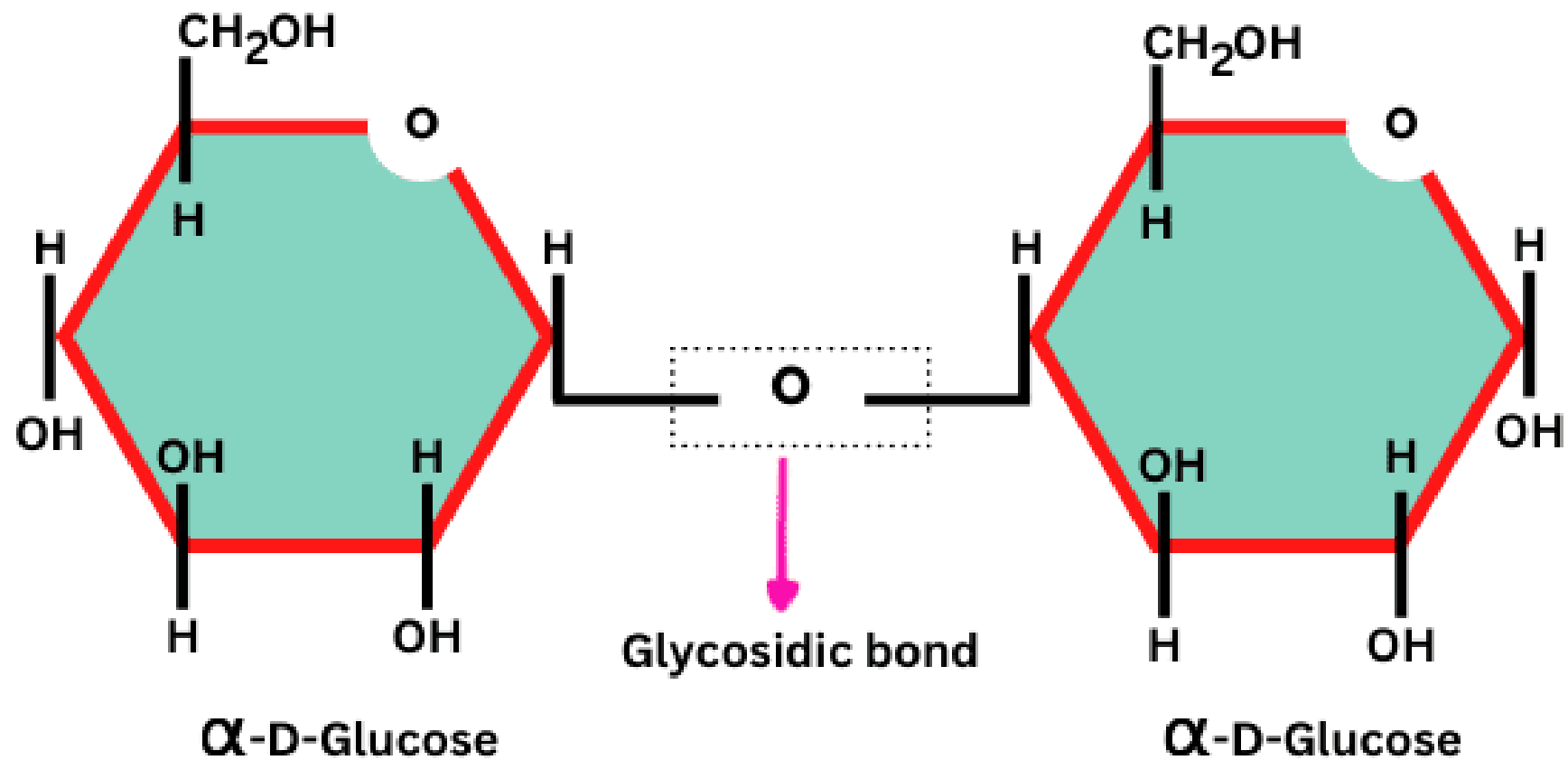
Glycoside Formation

- It involves the reaction of the **anomeric hydroxyl group** of a cyclic monosaccharide with an alcohol (or, more commonly in biology, with a hydroxyl group from another sugar molecule), with the elimination of a water molecule.
- The resulting bond formed is called a **glycosidic bond**.

Example: The reaction of the α -OH of glucose with ethanol forms ethyl- α -D-glucoside.

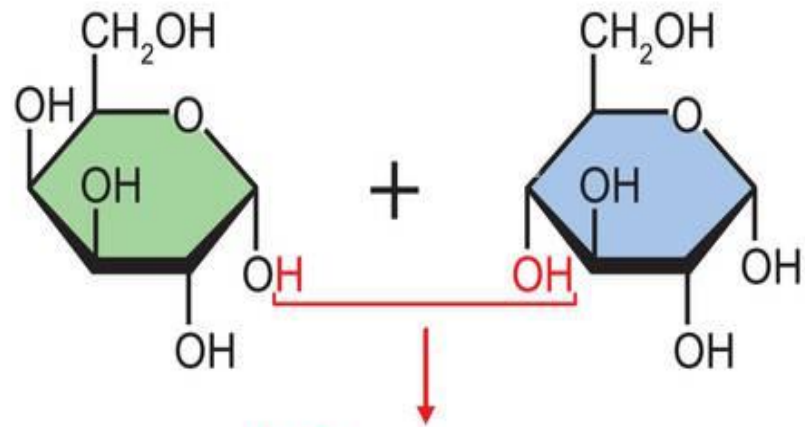
- **Biological Significance:** This reaction is the fundamental mechanism by which monosaccharides are linked together to form disaccharides, oligosaccharides, and polysaccharides. The specific configuration (alpha or beta) and the carbons involved (e.g., 1 $\rightarrow$ 4, 1 $\rightarrow$ 6) define the overall structure and properties of the larger carbohydrate.

A Glycosidic bond



Galactose

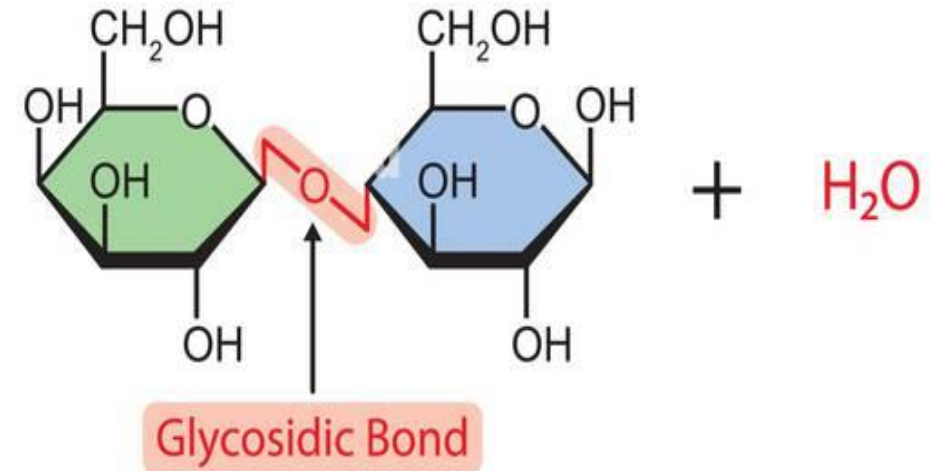
Glucose



Condensation

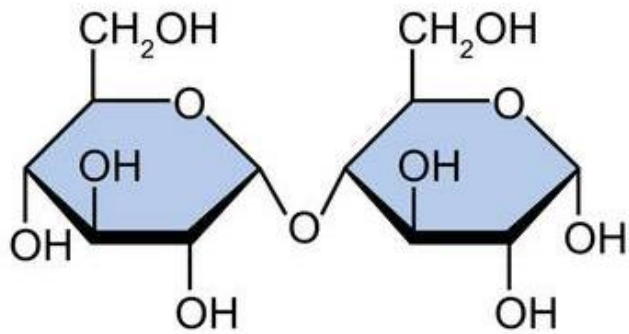


Lactose

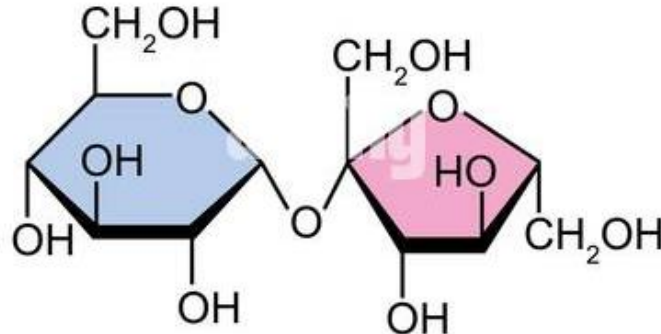


Disaccharides

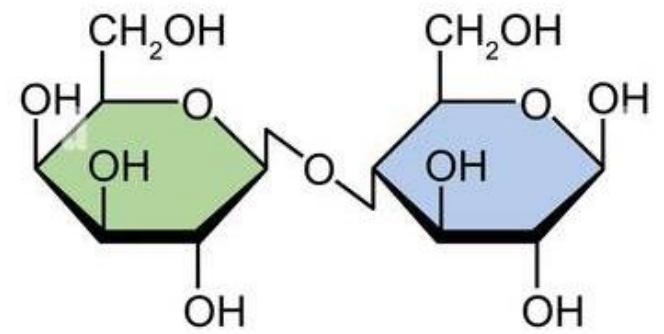
- Disaccharides are carbohydrates formed by the covalent linkage of two monosaccharide units via a **glycosidic bond**.
- This bond is formed through a dehydration reaction (loss of a water molecule).



Maltose



Sucrose



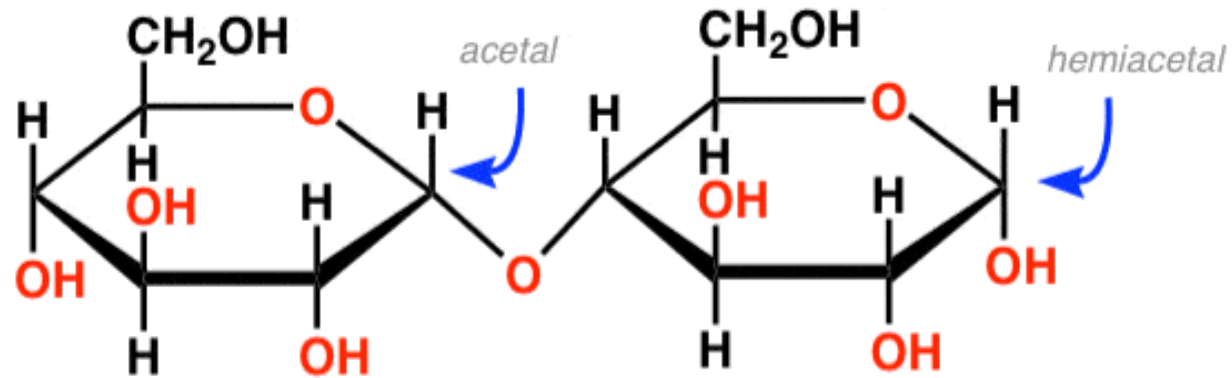
Lactose

Key Features of Disaccharides

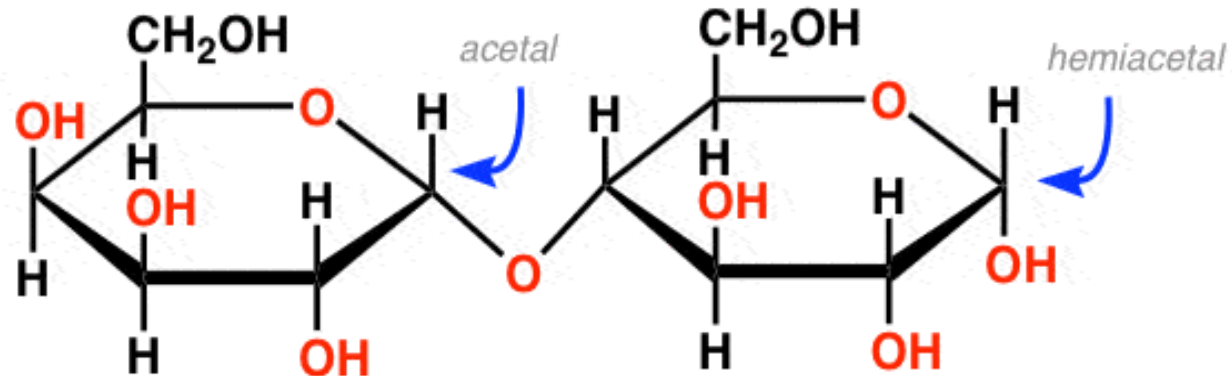
- They can be broken down (hydrolyzed) back into their constituent monosaccharides by either acid (non-specific) or specific enzymes (e.g., lactase hydrolyzes lactose).
- **Reducing vs. Non-reducing Disaccharides:** This classification depends on whether one of the anomeric carbons in the disaccharide remains "free" (not involved in the glycosidic bond) and therefore capable of opening to an aldehyde or ketone form.

Lactose and Maltose are also reducing sugars and give a positive Benedict test

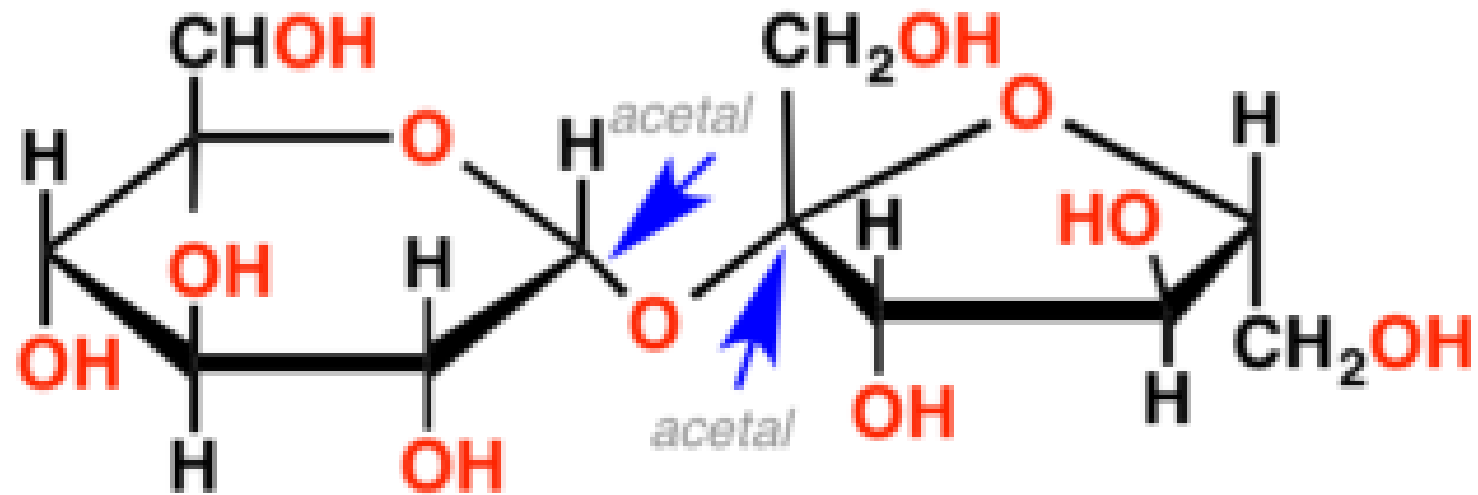
Maltose:



Lactose:



Sucrose is a non-reducing sugar



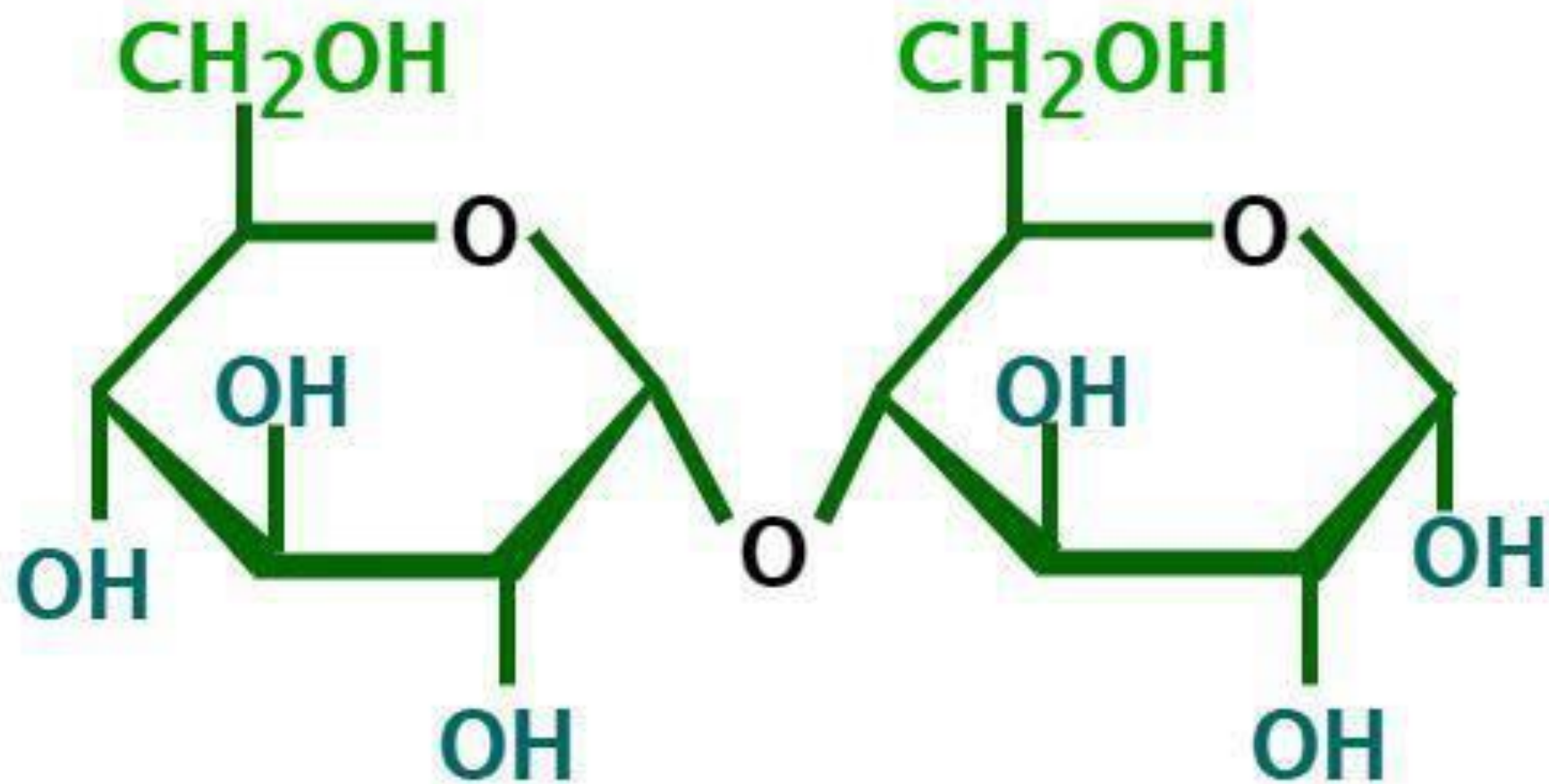
The structure lacks any hemiacetal functional groups and is therefore "locked" in its cyclic form

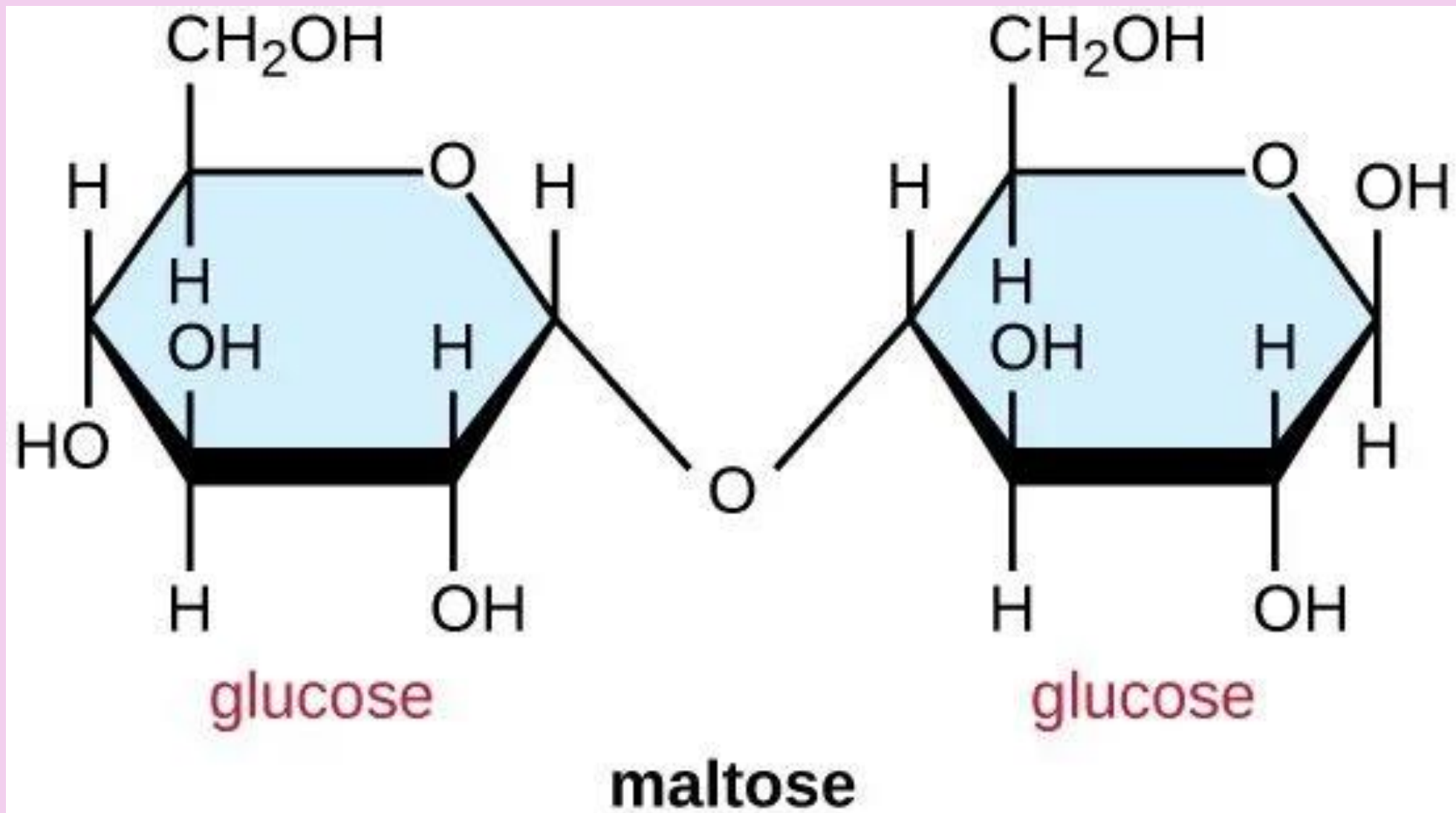


Disaccharides - Maltose (Malt Sugar)

- **Composition:** Two units of **Glucose** (alpha-D-glucose + D-glucose).
- **Glycosidic Linkage:** The C1 of one glucose unit (in the alpha-configuration) is linked to the C4 of the second glucose unit. ????
- **Reducing/Non-reducing:** Maltose is a **reducing sugar**. Although one anomeric carbon is involved in the glycosidic bond, the anomeric carbon of the *second* glucose unit is free to open into its aldehyde form.
- **Source:** It is a common intermediate product during the enzymatic hydrolysis (breakdown) of starch and glycogen (e.g., in germinating grains like barley, which is used in brewing).

Maltose





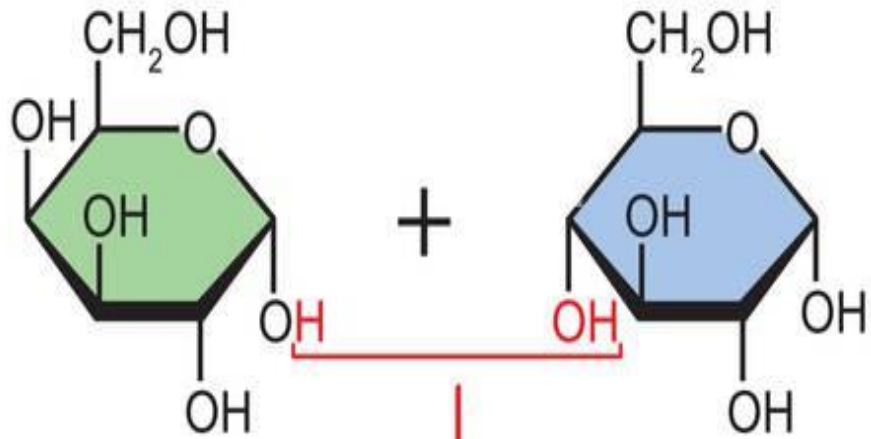
Disaccharides

Lactose (Milk Sugar)

- **Composition:** One unit of **Galactose** + one unit of **Glucose**.
- **Glycosidic Linkage:** The C1 of galactose (in the **beta-configuration**) is linked to the C4 of glucose. ????
- **Reducing/Non-reducing:** Lactose is a **reducing sugar**, as the anomeric carbon of the glucose unit is free.
- **Source:** Found almost exclusively in milk and dairy products.
- **Biological Significance:** The enzyme **lactase** (found in the small intestine) is required to hydrolyze lactose into glucose and galactose for absorption. Individuals with **lactose intolerance** lack sufficient lactase, leading to digestive discomfort when consuming dairy.

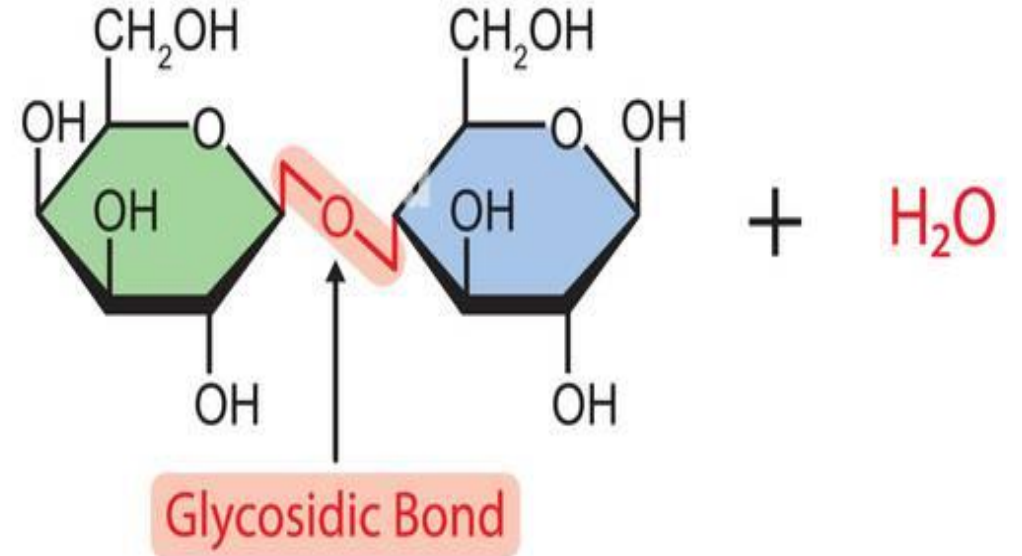
Galactose

Glucose



Condensation

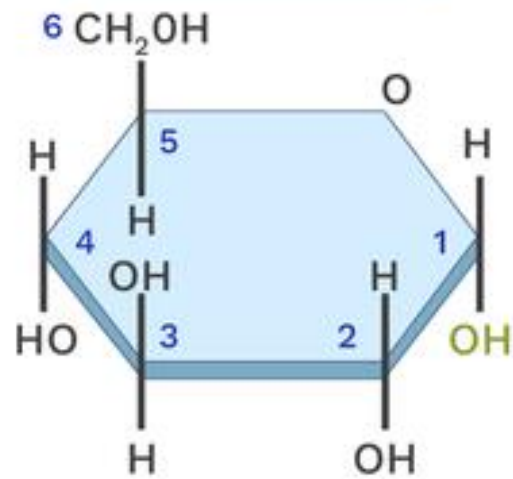
Lactose



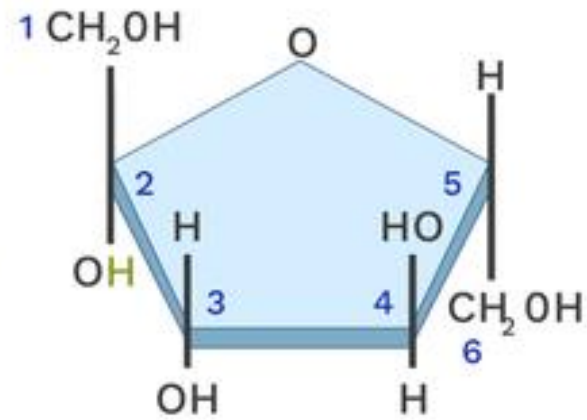
Disaccharides - Sucrose (Table Sugar / Cane Sugar)

- **Composition:** One unit of **Glucose** + one unit of **Fructose**.
- **Glycosidic Linkage:** This is a unique linkage! The anomeric carbon of glucose (C1, in the alpha-configuration) is linked to the anomeric carbon of fructose (C2, in the beta-configuration). ????
- **Reducing/Non-reducing:** Sucrose is a **non-reducing sugar**. Why? Because *both* anomeric carbons (C1 of glucose and C2 of fructose) are involved in the glycosidic bond. Neither can open up to a free aldehyde or ketone group.
- **Source:** Derived from sugarcane and sugar beets. It is the most abundant disaccharide in nature and is widely used as a sweetener.
- **Hydrolysis:** Upon acid hydrolysis or enzymatic hydrolysis by **sucrase** (also called invertase), sucrose yields an equimolar mixture of glucose and fructose. This mixture is often called "invert sugar" because the optical rotation changes from positive to negative during the process.

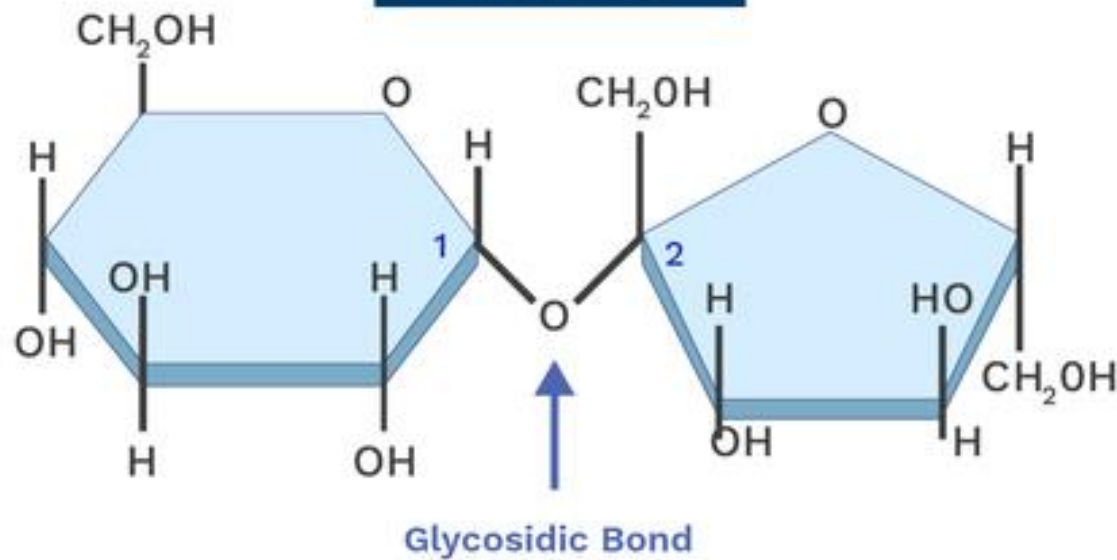
Glucose



Fructose



Sucrose



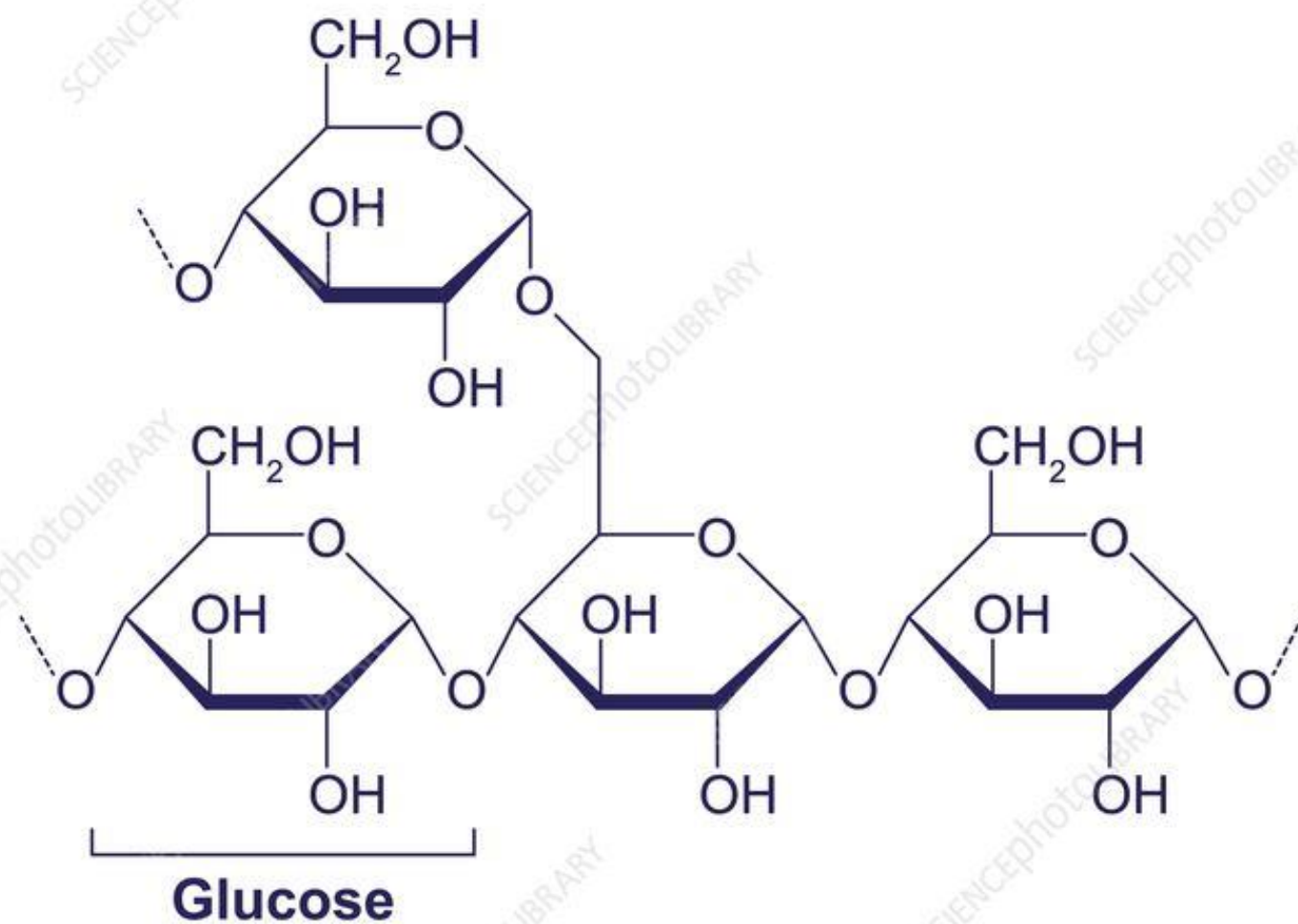
Polysaccharides

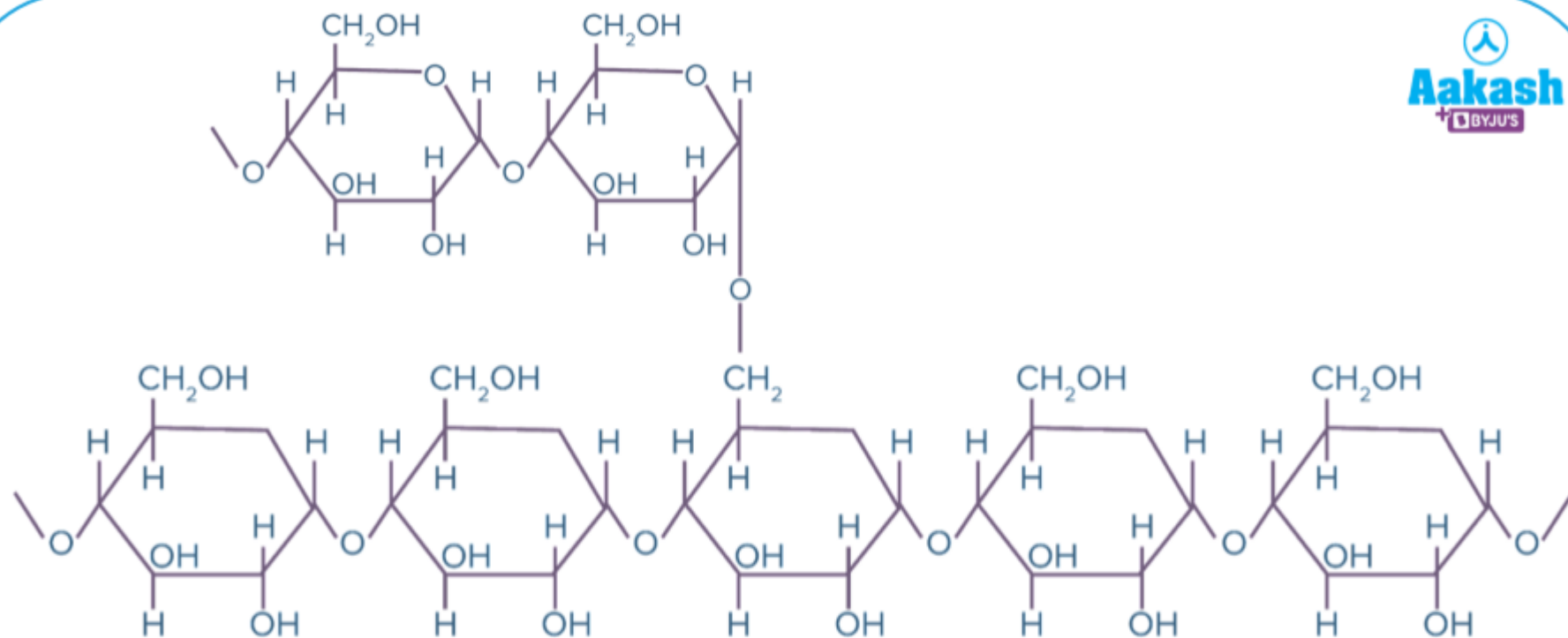
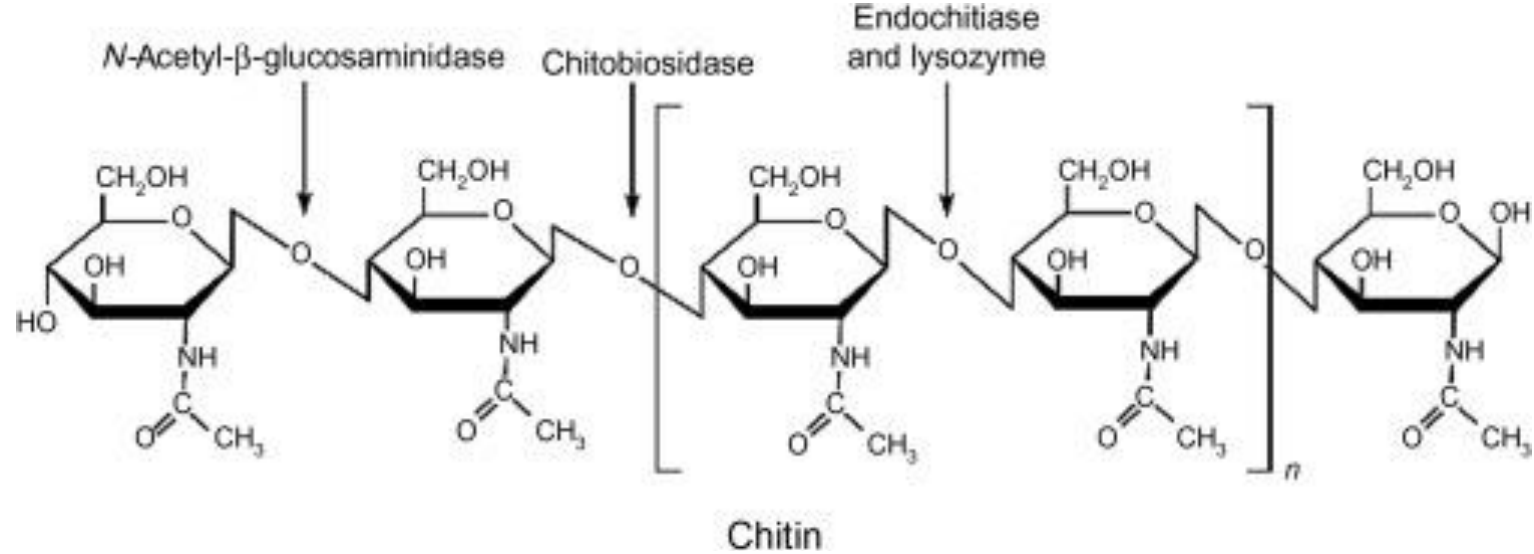
- Polysaccharides (also known as glycans) are very large polymers composed of many (hundreds to thousands) monosaccharide units linked together by glycosidic bonds.
- Their diverse structures, determined by the type of monosaccharide units, the type of glycosidic linkages (alpha or beta), and the degree of branching, lead to a wide range of biological functions.

Classification of Polysaccharides

- **Homopolysaccharides:** Composed of only *one* type of monosaccharide unit repeated many times (e.g., starch, glycogen, cellulose, chitin are all made of glucose or a glucose derivative).
- **Heteropolysaccharides:** Composed of *two or more different* types of monosaccharide units (e.g., peptidoglycans in bacterial cell walls, glycosaminoglycans in connective tissue).

Starch Structure



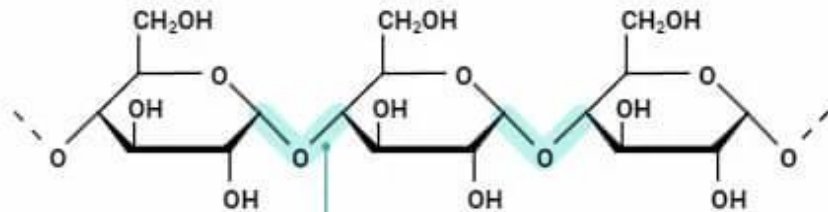


Structure of Glycogen

Storage Polysaccharides - Starch

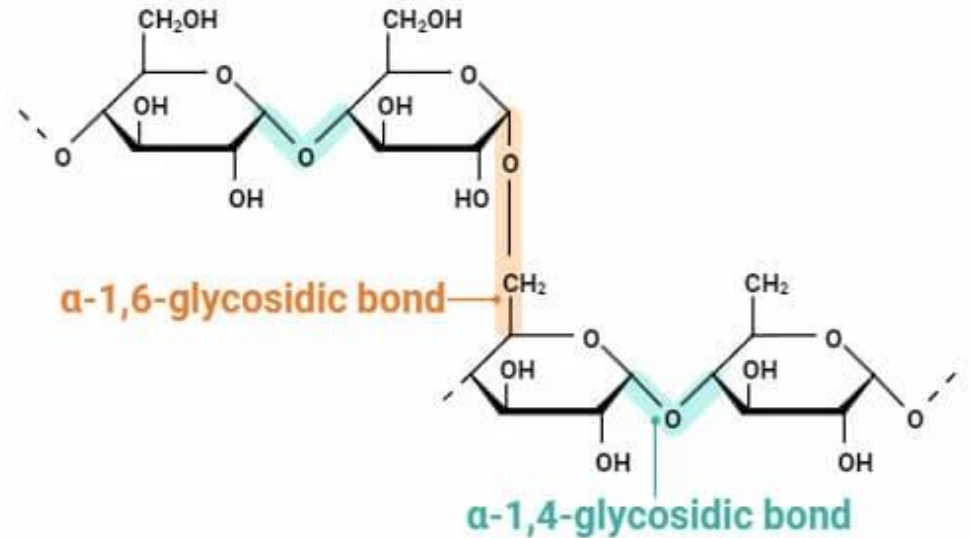
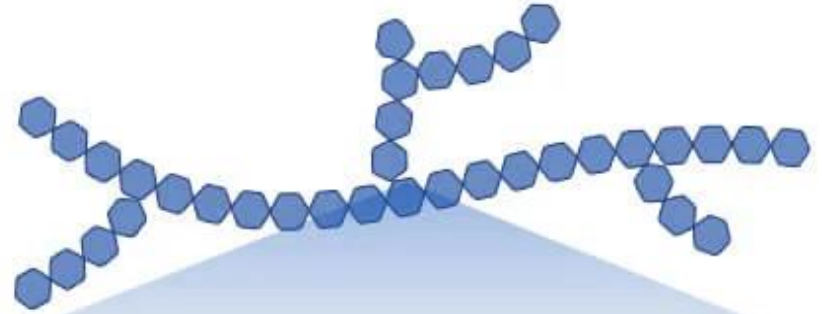
- **Function:** The principal energy storage polysaccharide in **plants**. It's how plants store excess glucose for later use.
- **Composition:** A homopolymer exclusively made of **glucose** units.
- **Two Major Components:**
 - **Amylose:** A linear (unbranched) polymer of glucose units linked predominantly by..... Due to these linkages, amylose typically adopts a helical (coiled) structure in solution.
 - **Amylopectin:** A branched polymer of glucose units. Its main chain consists oflinkages, but it has branches (typically every 24-30 glucose residues) formed by ?

Amylose



α -1,4-glycosidic bond

Amylopectin



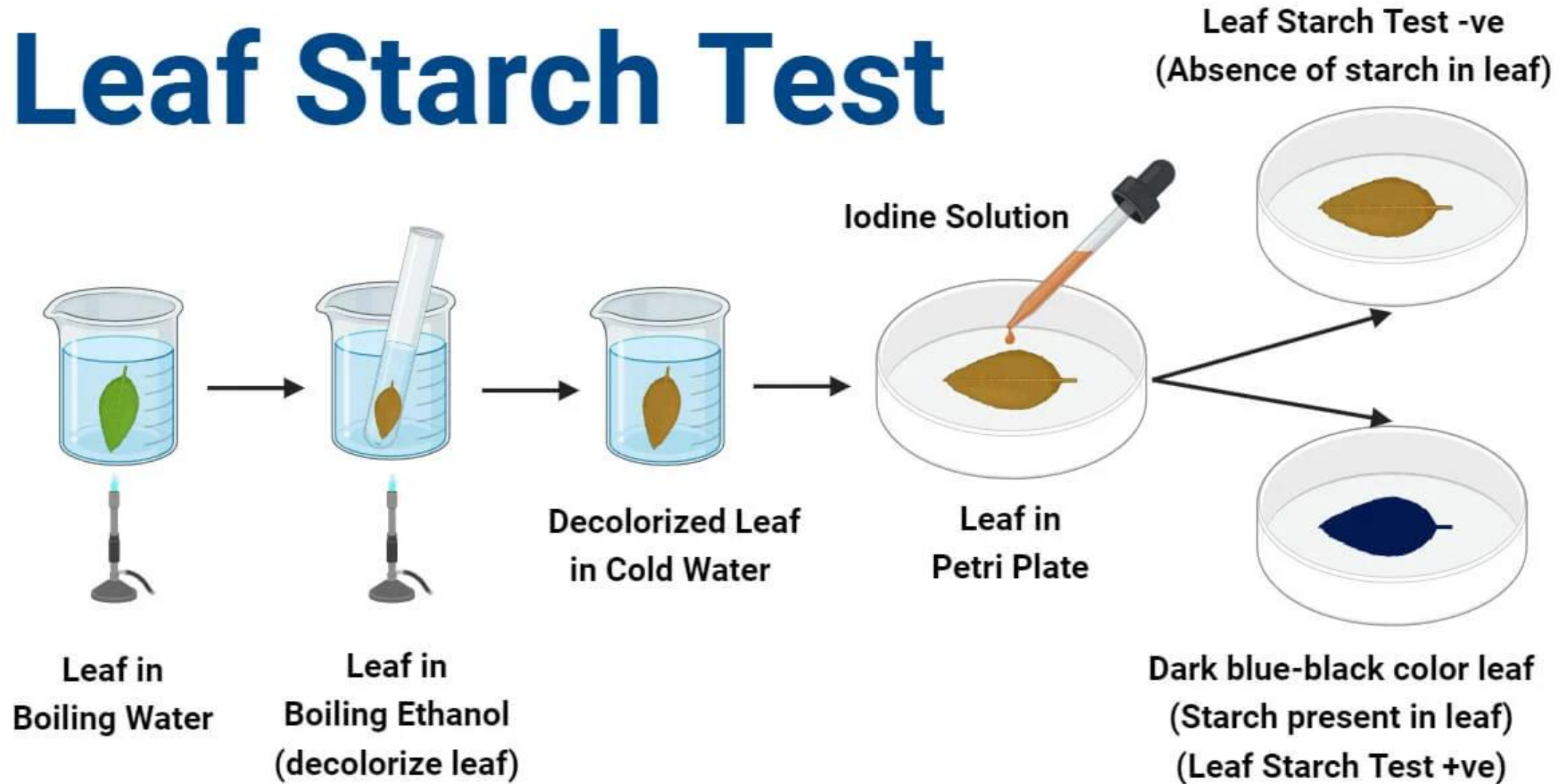
α -1,6-glycosidic bond

α -1,4-glycosidic bond

Storage Polysaccharides - Starch

- **Properties:** Starch is generally insoluble in water (amylose can form a colloidal suspension). When treated with iodine solution, starch (specifically the helical amylose) gives a characteristic **blue-black color**, which is a common diagnostic test.
- **Hydrolysis:** Starch is digested in our bodies by enzymes called **amylases** (e.g., salivary amylase, pancreatic amylase), which break the, yielding maltose and smaller oligosaccharides (dextrins), eventually being fully broken down to individual glucose units for absorption.

Leaf Starch Test

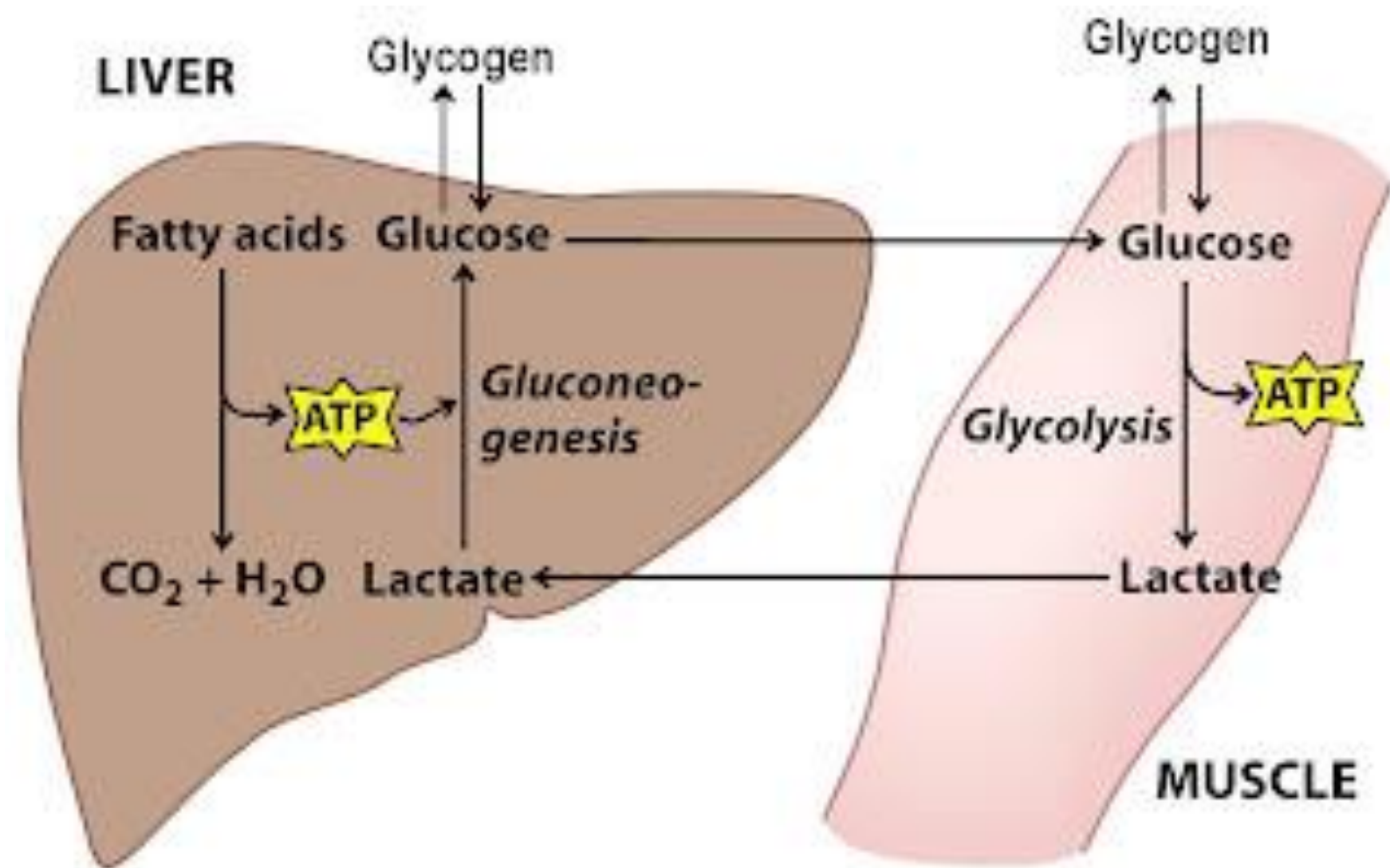


Storage Polysaccharides - Glycogen

- **Function:** The main energy storage polysaccharide in **animals** (including humans). It's primarily stored in the **liver** (for maintaining blood glucose levels) and **muscles** (for immediate energy during activity).
- **Composition:** A homopolymer made entirely of **glucose** units.
- **Structure:** Very similar to amylopectin but significantly *more highly branched* (branches occur every 8-12 glucose residues). This extensive branching gives it a very compact structure.

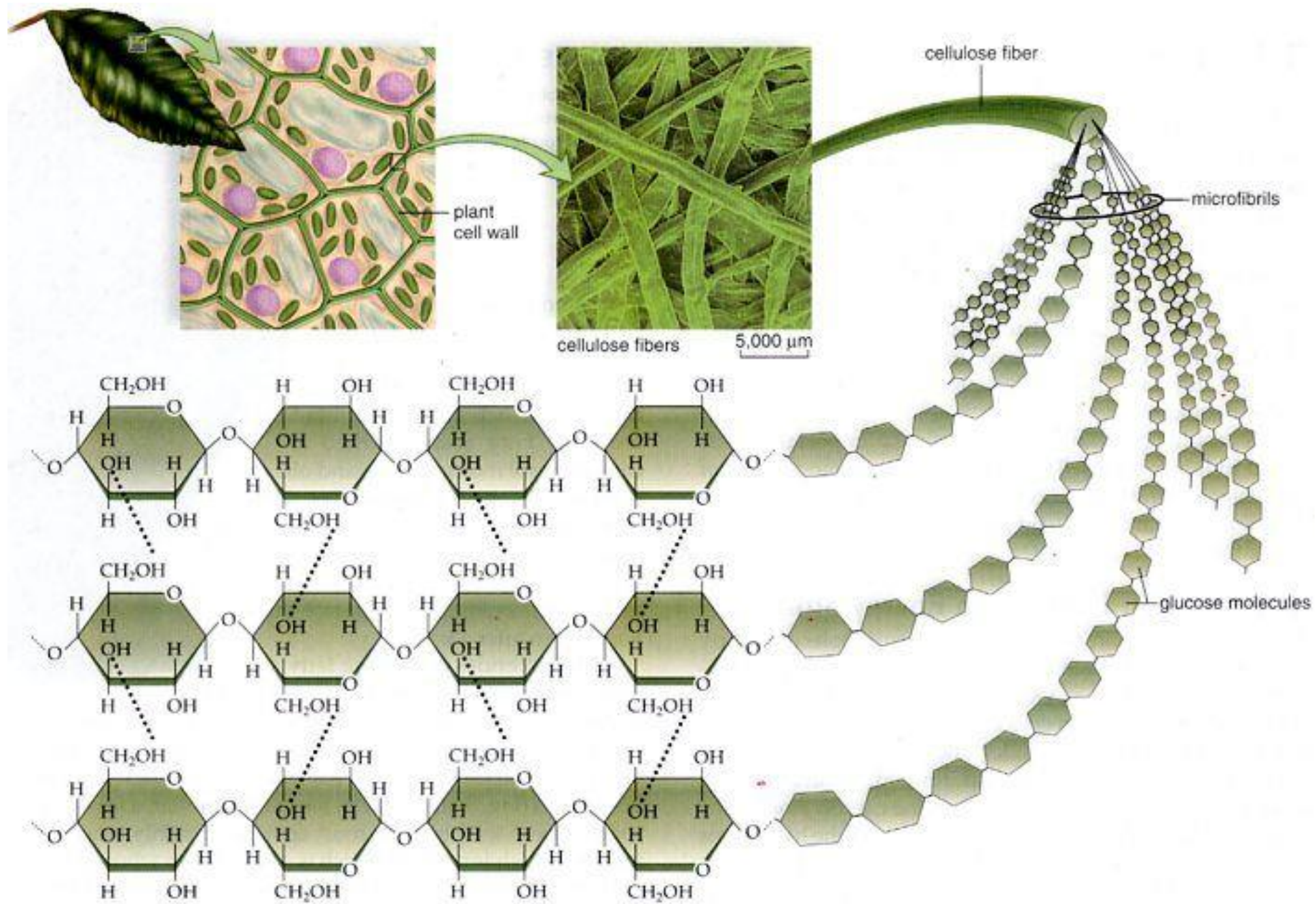
Storage Polysaccharides - Glycogen

- **Properties:** Highly soluble in water due to its compact and highly branched nature. The many non-reducing ends (the ends of the branches where glucose units can be removed) allow for very rapid mobilization of glucose when energy is needed.
- **Hydrolysis:** Digested by enzymes like **glycogen phosphorylase** (which removes glucose units from the non-reducing ends) and debranching enzymes, releasing glucose for energy.



Structural Polysaccharides - Cellulose

- **Function:** The primary structural component of **plant cell walls**. It provides rigidity, strength, and support to plants, allowing them to stand upright. It's the most abundant organic compound on Earth.
- **Composition:** A homopolymer made entirely of **glucose** units.
- **Glycosidic Linkage:** The key difference from starch and glycogen! Cellulose is linked by.....
- **Structure:** A long, linear (unbranched) chain. Because of the beta-linkages, each successive glucose unit is essentially flipped 180° relative to its neighbor. This specific orientation allows for extensive intra-chain and inter-chain **hydrogen bonding** between parallel cellulose chains, forming strong, insoluble **microfibrils**. These microfibrils bundle together to form even stronger fibers.



Structural Polysaccharides - Cellulose

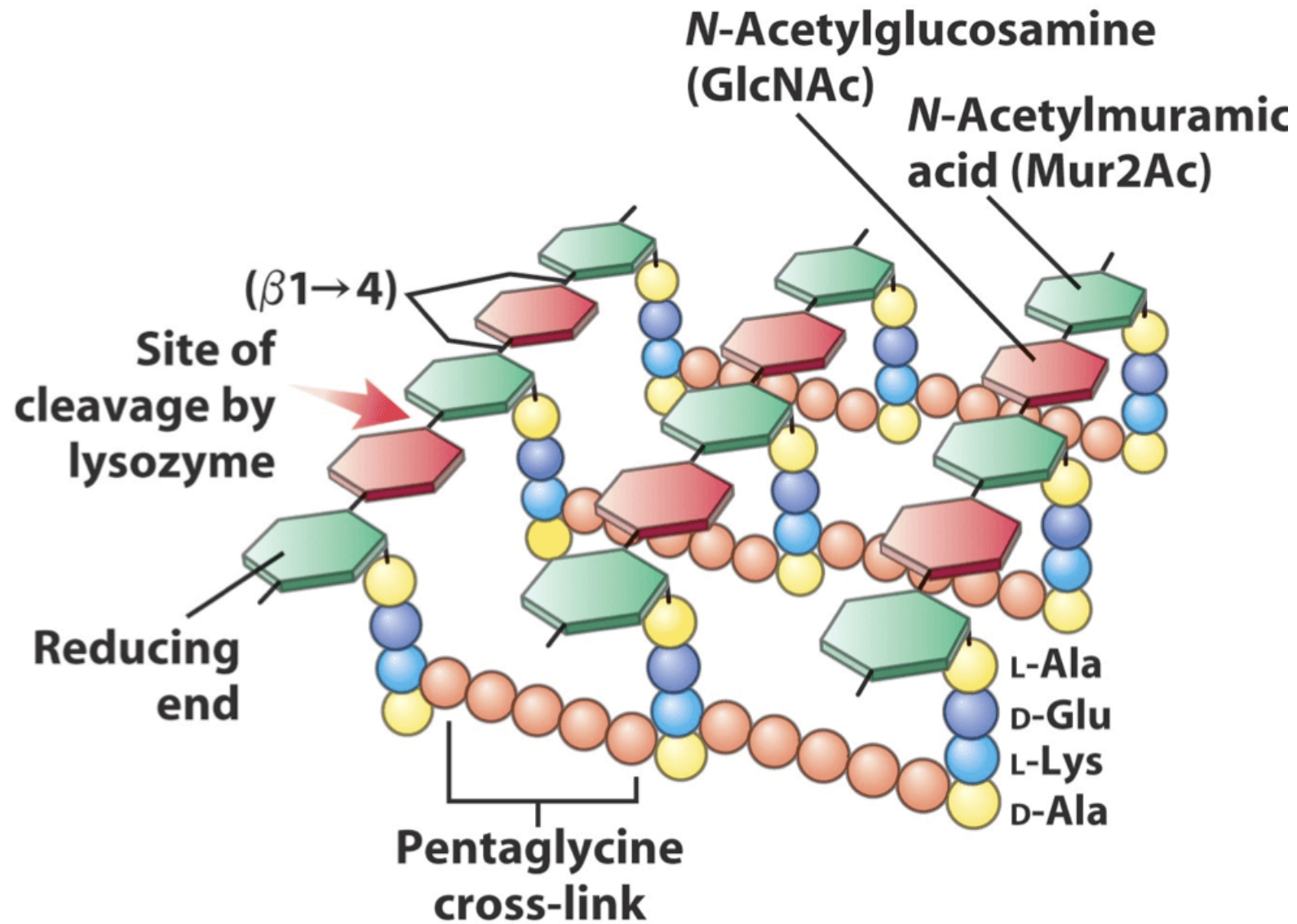
- **Properties:** Highly insoluble in water. Extremely stable and resistant to degradation.
- **Digestion:** Most animals, including humans, lack the enzyme **cellulase** which is required to hydrolyze theglycosidic bonds. Therefore, cellulose acts as **dietary fiber** (roughage) in our diets, promoting digestive health but providing no nutritional energy. Ruminant animals (like cows) and termites have symbiotic microorganisms in their digestive tracts that produce cellulase, allowing them to digest cellulose.

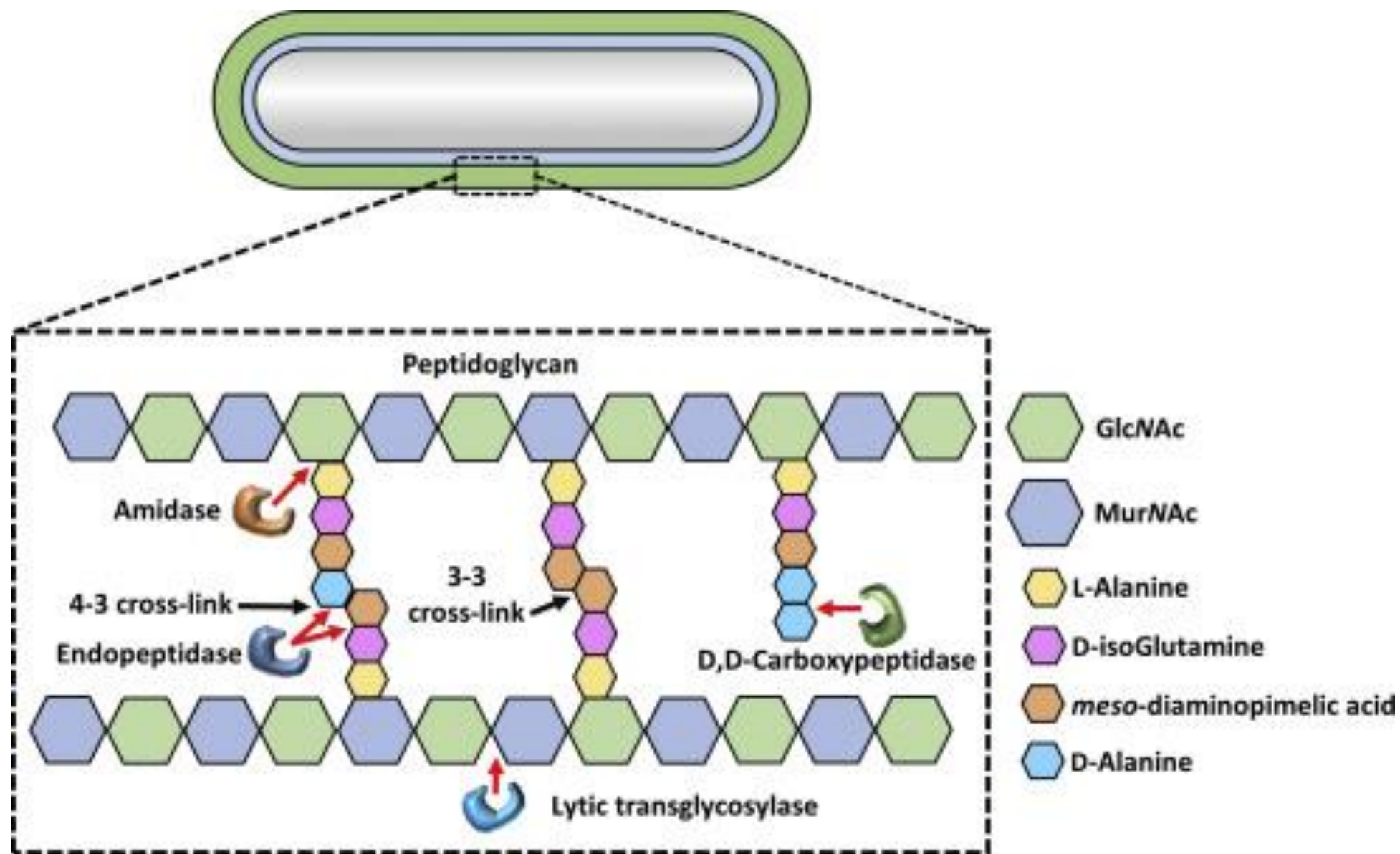
Structural Polysaccharides - Chitin

- **Function:** A major structural component of the **exoskeletons of insects and crustaceans** (e.g., crabs, lobsters), and the **cell walls of fungi**.
- **Composition:** A homopolymer of **N-acetylglucosamine** units. N-acetylglucosamine is a derivative of glucose where the hydroxyl group at C2 is replaced by an acetylated amino group.
- **Glycosidic Linkage:** Linked by....., similar to cellulose.
- **Structure:** Forms strong, linear fibrils due to extensive hydrogen bonding, similar to cellulose, providing tough and protective outer coverings.
- **Properties:** Insoluble, rigid, and highly durable.

Other Important Polysaccharides - Peptidoglycan

- **Function:** The major component of **bacterial cell walls**. Provides structural integrity and protects the bacterial cell from osmotic lysis (bursting).
- **Composition:** A heteropolysaccharide consisting of alternating units of **N-acetylglucosamine (NAG)** and **N-acetylmuramic acid (NAM)**, linked bybonds. These polysaccharide chains are further cross-linked by short **peptide chains**, forming a strong, mesh-like network around the bacterial cell.
- **Significance:** The unique structure of peptidoglycan, particularly its peptide cross-links, is a prime target for antibiotics like **penicillin**, which inhibit the enzymes responsible for forming these cross-links, thereby weakening the bacterial cell wall and leading to cell death.



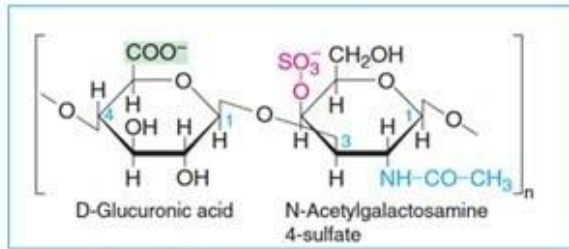


Other Important Polysaccharides - Glycosaminoglycans (GAGs) / Mucopolysaccharides

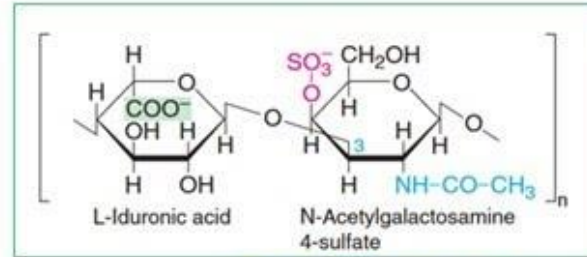
- **Function:** Crucial components of the **extracellular matrix (ECM)** in animals, the complex network surrounding cells. They provide lubrication, shock absorption, and structural support in tissues.
- **Composition:** Linear heteropolysaccharides made of repeating disaccharide units. Typically, one sugar in the disaccharide is an amino sugar (like N-acetylglucosamine or N-acetylgalactosamine), and the other is often a uronic acid (e.g., glucuronic acid or iduronic acid). GAGs are also often highly **sulfated** (contain sulfate groups).

Carbohydrates 20

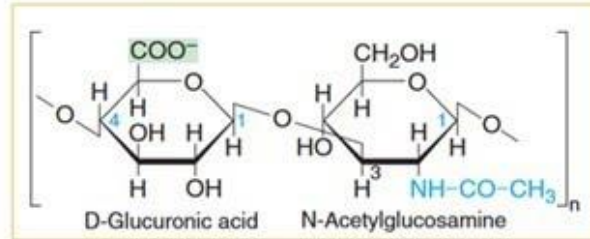
MUCOPOLYSACCHARIDES



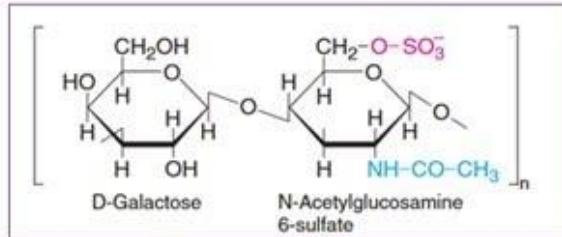
Chondroitin sulphate



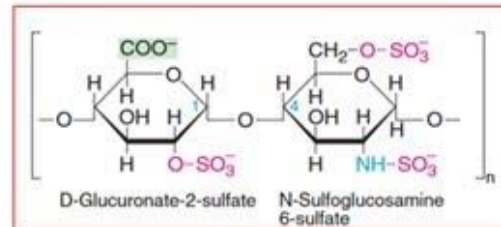
Dermatan sulphate



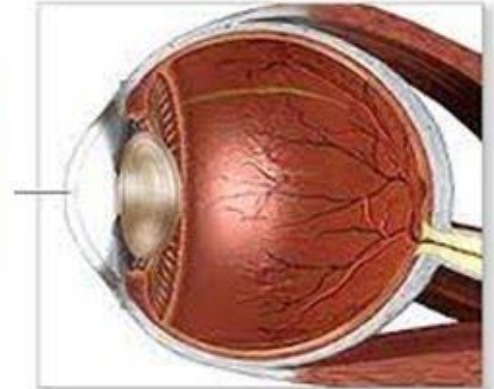
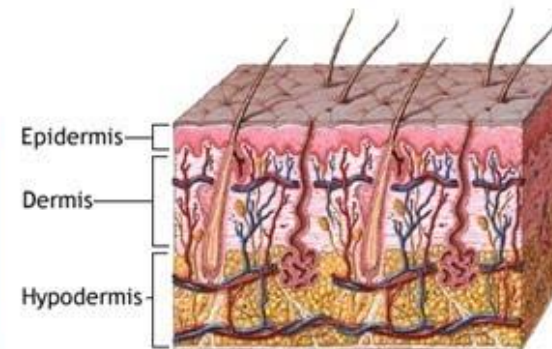
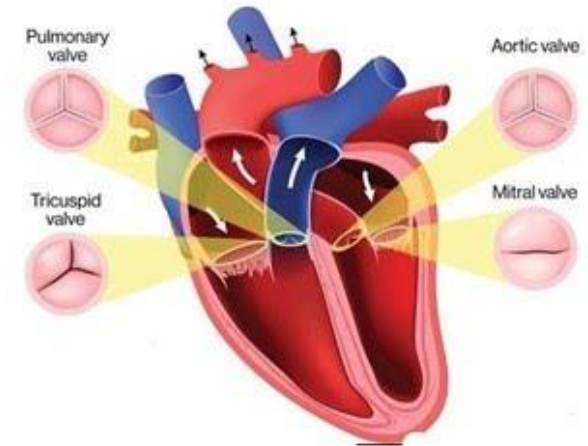
Hyaluronic acid



Keratan sulphate



Heparin



Other Important Polysaccharides - Glycosaminoglycans (GAGs) / Mucopolysaccharides

Examples:

- **Hyaluronic Acid:** Found in synovial fluid (lubricant in joints), the vitreous humor of the eye, and embryonic tissues. Unique among GAGs as it is unsulfated.
 - **Chondroitin Sulfate:** A major component of cartilage, bone, and skin, providing resistance to compression.
 - **Heparin:** A natural anticoagulant (prevents blood clotting) found in mast cells.
 - **Keratan Sulfate:** Found in cartilage, bone, and the cornea of the eye.
-
- **Properties:** The numerous negatively charged sulfate and carboxyl groups (from uronic acids) make GAGs highly polyanionic. This strong negative charge attracts large amounts of water, forming highly hydrated, viscous gels that can withstand compressive forces and provide lubrication.